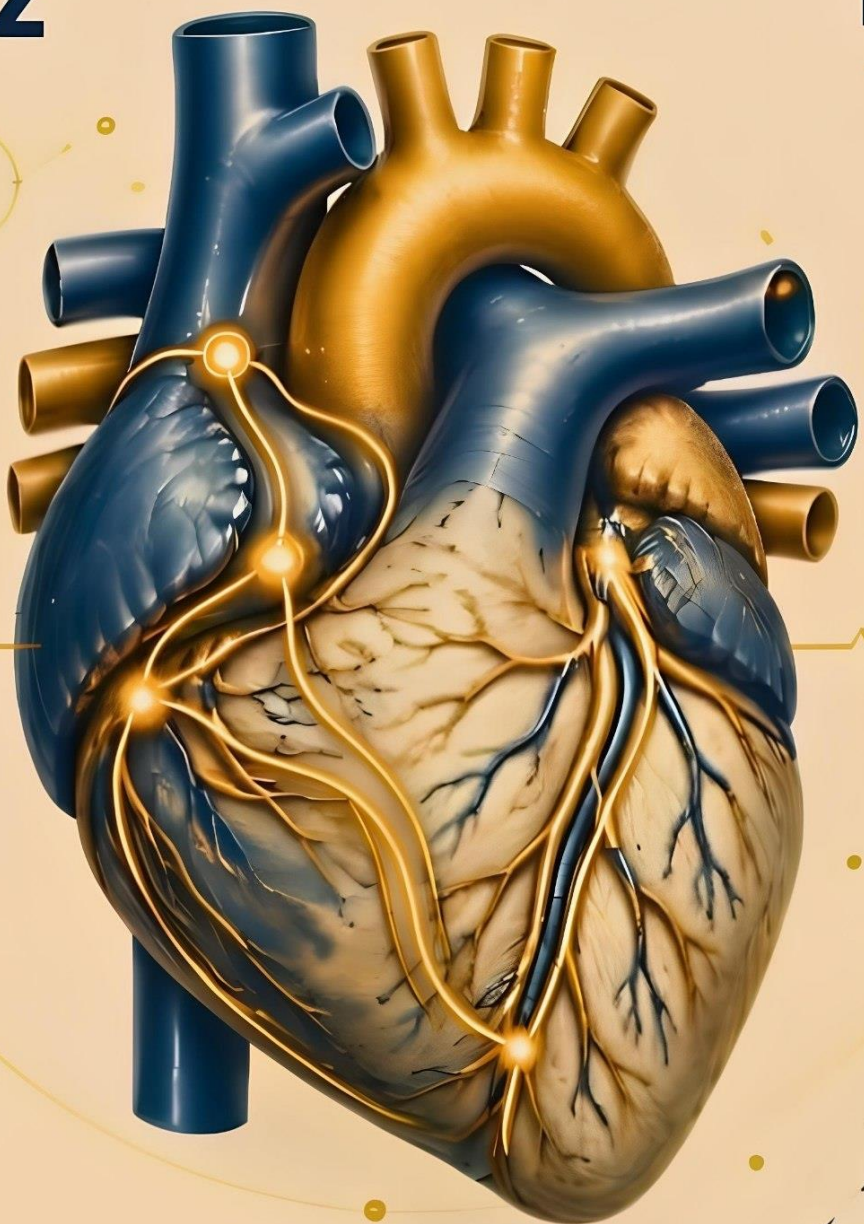
PHYSIOLOGY

Part 2

MCQS



2026

بسم الله الرحمن الرحيم

والصلاة والسلام على خير المرسلين محمد بن عبد الله وعلى آله وصحبه وسلم أما بعد، نقدم بين يديك ملف

(General Physiology MCQ)

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ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تمل

حسابات الفريق على منصات التواصل الاجتماعي

جميع الحقوق محفوظة لفريق مِعوان

ولا نحلل بيعها بعائد ربحي أو سرقتها بأي شكل من الأشكال

ANS II (Sympathetic nervous system)

Pre Exams & Formative

1-	**The sympathetic response ina "fight or flight" reaction causes decrease in? (MID 31, Formative 36)**		
A-	The arterial blood pressure	B-	The diameter of the pupil
C-	The resistance of the airways	D-	The blood glucose concentration

2-	**By which route are the sweat glands supplied with parasympathetic innervation? (MID 31)**
A-	Vagal preganglionics to paravertebral ganglion to cutaneous nerve
B-	Vagal preganglionics to prevertebral ganglion to cutaneous nerve
C-	Spinal preganglionics to paravertebral ganglion to cutaneous nerve
D-	There is no parasympathetic innervation to the sweat glands

3-	**Sympathetic stimulation? (MID 31)**		
A-	Results in increase cutaneous blood flow	B-	Indirectly increase the coronary blood flow
C-	Results in decrease muscle blood flow	D-	Causes vasodilatation in all vascular beds

4-	**Which of the following structure is innervated by preganglionic sympathetic nerve fibers? (MID 31, MID 32, 2ND TRIAL 32, Formative 36)**						
A-	Lung	B-	Heart	C-	Pancreas	D-	Adrenal medulla

5-	**A patient with Horner's syndrome may have lesion in? (FINAL 31)**		
A-	Superior cervical ganglion	B-	Inferior mesenteric ganglion
C-	Ciliary ganglion	D-	None of the above is correct

6-	**Concerning the sympathetic nervous system: (MID 32)**
A-	consists only of cranial nerves
B-	Its efferent pathway is made of one neuron
C-	All post ganglionic sympathetic fibers secrete NE
D-	Its stimulation causes dilation of the airways

7-	**Which of following fibers release NE? (FINAL 31)**
A-	Preganglionic fibers innervating the adrenal medulla
B-	Postganglionic fibers causing bradycardia
C-	Postganglionic fibers causing constriction of the iris
D-	Postganglionic fibers causing arteriolar constriction

8-	**As regard orbelli phenomenon all of the following is true EXCEPT? (FINAL 31)**
A-	Better contraction of skeletal muscle
B-	Delayed fatigue of skeletal muscle
C-	Increased blood supply to skeletal muscle
D-	Caused by parasympathetic stimulation to skeletal muscle

9-	**Stimulation of the sympathetic nervous system causes? (2ND TRIAL 32)**
A-	bronchoconstriction
B-	constriction of the pupil
C-	increases the heart rate & the force of ventricle contraction
D-	increases the heart rate but has no effect on the force of contraction

10-	**Stimulation of the greater splanchnic nerve produces? (MID 33)**
A-	Increased motility of the stomach & upper half of the small intestine
B-	Increased blood levels of both glucose & catecholamines
C-	V.C. of the blood vessels in the erectile tissue of males
D-	Increased secretion by both the stomach & pancreas

11-	**Postganglionic sympathetic vasodilator fibers? (FINAL 35)**
A-	Vasodilate the renal blood vessels
B-	Cause vasoconstriction of cerebral blood vessels
C-	Cause vasodilatation of pulmonary blood vessels
D-	Vasodilate the blood vessels of skeletal muscles

12-	**Centeral cholinergic sympathetic fibers (FINAL 37)**
A-	All post ganglia parasympathetic
B-	Sympathetic post ganglia that supply eccrine sweat gland
C-	Adrenal medulla

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
C	D	B	D	A	D	D	D	C	B
11	12	13	14	15	16	17	18	19	20
D	C	-	-	-	-	-	-	-	-

Department Qs

1-	All of the following about the sympathetic N.S. is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)
A-	All its pregang. fibres are cholinergic (secreting Ach.).
B-	It is involved in stress and emergency conditions.
C-	It has a V.D. effect on skeletal muscle blood vessels during rest.
D-	Its stimulation increases the cardiac properties.

2-	Concerning adrenal medulla, all the following is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)
A-	It acts only in emergency conditions.
B-	It secretes epinephrine 80% and norepinephrine 20%.
C-	It isa modified sympathetic ganglion.
D-	It receives preganglionic nerve supply from the anterior born cells of the lower 6 thoracic segments of the spinal cord.

3-	The sympathetic supply cannot be responsible for: (SNU Bank, Department Qs, Dr.Seham)		
A-	Constriction of the pupil.	B-	Erection of hair.
C-	Secretion of sweat gland.	D-	Dilatation of skeletal muscle blood vessels.

4-	The preganglionic sympathetic nerve that innervate the eye originate from: (SNU Bank, Department Qs)		
A-	Superior colliculi.	B-	Thoracic segments of spinal cord.
C-	Superior cervical ganglion.	D-	Ciliary ganglion.

5-	Sympathetic supply to the pelvic viscera cause: (SNU Bank, Department Qs, Dr.Seham)		
A-	Micturation.	B-	Dilatation of vessels in external genetalia.
C-	Defecation.	D-	Ejaculation.

6-	Postganglionic sympathetic starting from cervical and upper four thoracic ganglia causes: (SNU Bank, Department Qs, Dr.Seham)		
A-	Secretion of the adrenomedullary hormones.	B-	Dilatation of the coronary vessels.
C-	Contraction of the splenic capsule.	D-	Narrowing of the respiratory passages.

7-	Increased activity of the sympathetic nervous system causes: (SNU Bank, Department Qs, Dr.Seham)		
A-	Penile erection.	B-	Pupillary constriction.
C-	Accommodation for near vision.	D-	Bronchiolar dilatation.

8-	**A lesion in the cervical symp. chain (Horner's syndrome) causes : (SNU Bank, Department Qs, Dr.Seham)**
A-	Increased sweating.
B-	Dropping of the upper eyelid (=ptosis).
C-	Vasoconstriction of the skin blood vessels (leading to pallor).
D-	Mydriasis dilatation of the pupil).

9-	**Stimulation of the greater splanchnic nerve produces : (SNU Bank, Department Qs, Dr.Seham)**
A-	Increased motility of the plain muscles of the stomach.
B-	Excessive secretion of the gastric juice by the stomach.
C-	Decreased blood levels of both glucose & free fatty acids.
D-	Secretion of epinephrine & norepinephrine by the adrenal medulla.

10-	**Stimulation of the sympathetic N.S. causes : (SNU Bank)**
A-	Contraction of the ciliary muscle for near vision.
B-	Generalized actions affecting many systems.
C-	Bronchoconstriction.
D-	Decreased glycogenolysis & lipolysis.

11-	**Stimulation of the cervical sympathetic nerve causes : (SNU Bank, Department Qs, Dr.Seham)**		
A-	Dropping of the upper eyelid.	B-	A small-sized pupil.
C-	Enophthalmos.	D-	Relaxation of the ciliary muscle of the eye.

12-	**Stimulation of the greater splanchnic nerve produces: (SNU Bank)**
A-	Increased motility of the stomach & upper half of the small intestine
B-	Inhibition of micturition
C-	Increased blood level of both glucose & catecholamines
D-	V'C. of the blood vessels in the erectile tissue of both males & females

13-	**Stimulation of lesser splanchnic nerve causes: (SNU Bank, Department Qs, Dr.Seham)**						
A-	Micturition.	B-	Defecation.	C-	Ejaculation.	D-	Erection.

14-	**Stimulation of greater Splanchnic nerve produces: (SNU Bank, Department Qs, Dr.Seham)**
A-	Increased blood glucose level.
B-	Decreased blood volume.
C-	Secretion of epinephrine 20% & norepinephrine 80% from adrenal medulla.
D-	Increased gastrointestinal motility.

15-	**Activation of sympathetic nervous system produces the following EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	Bronchodilatation.
B-	Tachycardia.
C-	Mydriasis (dilatation of pupil).
D-	Contraction of detrusor muscle of urinary bladder wall.

16-	**All of the following about Horner syndrome is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	Manifested by dryness & redness of skin.
B-	Group of signs including ptosis & miosis.
C-	Is due to injury ofL'H'Cs. present in thoracic 1-2 spinal segments.
D-	Causes pallor of skin due toV'C.

17-	**Choose the correct statement: (SNU Bank, Department Qs, Dr.Seham)**
A-	Postganglionics starting from the superior cervical ganglion causes constriction of the pupils.
B-	Sympathetic nervous system supplies every organ in the body
C-	Sympathetic supply to the lungs causes bronchoconstriction.
D-	Stimulation of greater splanchnic nerve results in adrenomedullary secretion.

18-	**Activation of sympathetic system causes: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Pupillary constriction.	B-	Sweat secretion.
C-	Bradycardia.	D-	Cutaneous vasodilatation.

19-	**Among symptoms of horner's syndrome there is: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Increased sweating.	B-	Dilatation of pupil.
C-	Dropping of the upper eye lid.	D-	Vasoconstriction of skin blood vessels.

20-	**Generalized sympathetic stimulation causes: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Inhibition of sweating.	B-	Increase gastric motility.
C-	Decreases blood sugar levels.	D-	Rise arterial blood pressure.

21-	**Horner's syndrome - lesion in? (SNU Bank)**		
A-	Superior cervical ganglion	B-	Inferior mesenteric ganglion
C-	Ciliary ganglion	D-	None of the above

22-	**Orbelli phenomenon - all true EXCEPT? (SNU Bank)**		
A-	Better contraction of skeletal muscle	B-	Delayed fatigue of skeletal muscle
C-	Increased blood supply to skeletal muscle	D-	Caused by parasympathetic stimulation

23-	Which of the following structure is innervated by preganglionic sympathetic nerve fibers? (SNU Bank)				
A-	Lung	B-	Heart	C-	Pancreas
D-	Adrenal medulla	E-	Skeletal muscle		

24-	Stimulation of the sympathetic nervous system causes? (SNU Bank)
A-	Bronchoconstriction
B-	Constriction of the pupil
C-	Increases the heart rate & the force of ventricle contraction
D-	Increases the heart rate but has no effect on the force of contraction

25-	Stimulation of greater splanchnic nerve produces all the following EXCEPT: (SNU Bank, Dr.Seham)
A-	Secretion of catecholamines from the suprarenal medulla
B-	Ejaculation
C-	Increased blood glucose level
D-	Decreased gastrointestinal motility

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	D	A	B	D	B	D	B	D	B
11	12	13	14	15	16	17	18	19	20
D	C	C	A	D	D	D	B	C	D
21	22	23	24	25	26	27	28	29	30
A	D	D	C	B	-	-	-	-	-

M3WAN Bank

1-	The sympathetic nervous system is primarily concerned with body activities during: (M3WAN - Core)
A-	Digestion, Defecation, and Diuresis (the 'D' division)
B-	Rest and sleep only
C-	Growth and development exclusively
D-	Exercise, Excitement, Emergency, and Embarrassment (the 'E' division, 'fight or flight')

2-	The sympathetic nervous system can be divided into how many main divisions, and what are they? (M3WAN - Core)
A-	Two divisions: cranial and sacral
B-	Three divisions: thoracic, lumbar, and sacral
C-	Four divisions: cervical, cardio-pulmonary, splanchnic, and somatic
D-	Five divisions, one for each spinal region

3-	**The cervical sympathetic division mainly supplies structures in the region of: (M3WAN - Core)**		
A-	The heart and lungs	B-	The head and neck
C-	The abdominal and pelvic viscera	D-	The skeletal muscles only

4-	**The cervical sympathetic division arises from the lateral horn cells of: (Important M3WAN - Core)**
A-	The upper 2 thoracic segments of the spinal cord
B-	The upper 4 thoracic segments of the spinal cord
C-	The lower 6-7 thoracic and upper 3 lumbar segments
D-	The 2nd, 3rd, and 4th sacral segments

5-	**Preganglionic fibers of the cervical sympathetic division relay in the: (M3WAN - Core)**		
A-	Celiac ganglion	B-	Inferior mesenteric ganglion
C-	Superior cervical ganglion	D-	Ciliary ganglion

6-	**Sympathetic stimulation of the eye causes contraction of the dilator pupillae muscle, leading to: (Important M3WAN - Core)**		
A-	Miosis (pupil constriction)	B-	Accommodation for near vision
C-	Complete closure of the eyelid	D-	Mydriasis (pupil dilation)

7-	**Sympathetic stimulation of the eye causes contraction of the superior and inferior tarsal muscles, leading to: (M3WAN - Core)**		
A-	Narrowing of the palpebral fissure	B-	Widening of the palpebral fissure
C-	Closure of the eyelids	D-	Excessive tear secretion

8-	**Sympathetic stimulation causes contraction of Muller's muscle behind the eyeball, leading to (especially in animals): (M3WAN - Hard)**						
A-	Exophthalmos	B-	Enophthalmos	C-	Ptosis	D-	Miosis

9-	**Sympathetic stimulation of the blood vessels of the eye causes: (M3WAN - Core)**		
A-	Vasodilation	B-	No change in vessel diameter
C-	Alternating constriction and dilation	D-	Vasoconstriction

10-	**Sympathetic stimulation causes relaxation of the ciliary muscle, which is used mainly for: (M3WAN - Core)**		
A-	Near vision (accommodation)	B-	Color perception
C-	Far vision	D-	Dark adaptation

11-	**Sympathetic stimulation of the salivary glands causes vasoconstriction leading to: (M3WAN - Core)**
A-	True secretion: profuse and watery, with little enzyme content
B-	Trophic secretion: little in amount, viscid, and rich in enzymes

C-	Complete cessation of all salivary secretion
D-	A doubling of salivary flow rate

12-	**Sympathetic stimulation of the lacrimal glands produces: (M3WAN - Core)**		
A-	Vasoconstriction with a little trophic secretion	B-	Vasodilation with profuse watery secretion
C-	Complete inhibition of any secretion	D-	Only sensory changes with no secretory effect

13-	**Sympathetic stimulation of the skin of the head and neck causes all of the following EXCEPT: (M3WAN - Hard)**
A-	Vasoconstriction
B-	Sweat secretion
C-	Hair erection
D-	Relaxation of the erector pilae (pilomotor) muscles

14-	**Sympathetic stimulation of the cerebral blood vessels causes: (M3WAN - Core)**		
A-	Strong vasodilation	B-	No effect whatsoever
C-	Weak vasoconstriction	D-	Complete vascular occlusion

15-	**Horner's syndrome results from interruption of the sympathetic supply to the head and neck, typically due toa lesion in: (Important M3WAN - Core)**
A-	The first and second thoracic segments or the superior cervical ganglion
B-	The sacral segments of the spinal cord
C-	The vagus nerve nucleus
D-	The oculomotor nucleus

16-	**In Horner's syndrome, drooping of the upper eyelid on the affected side is called: (M3WAN - Core)**						
A-	Miosis	B-	Ptosis	C-	Anhidrosis	D-	Enophthalmos

17-	**In Horner's syndrome a smaller than normal pupil in the affected eye is called: (Important M3WAN - Core)**						
A-	Mydriasis	B-	Ptosis	C-	Exophthalmos	D-	Miosis

18-	**In Horner's syndrome, lack of sweating on the affected side of the face is called: (M3WAN - Core)**						
A-	Hyperhidrosis	B-	Ptosis	C-	Anhidrosis	D-	Miosis

19-	**In Horner's syndrome, sinking of the eye into its socket is called: (M3WAN - Core)**						
A-	Exophthalmos	B-	Enophthalmos	C-	Ptosis	D-	Anhidrosis

20-	In congenital cases of Horner's syndrome,a lighter iris color in the affected eye compared with the unaffected eye is called: (M3WAN - Hard)		
A-	Iris heterochromia	B-	Iris coloboma
C-	Anisocoria alone, without color change	D-	Heterophoria

21-	The cardio-pulmonary sympathetic division, which supplies the heart and lungs, arises from the lateral horn cells of: (M3WAN - Core)
A-	The upper 2 thoracic segments
B-	The lower 6-7 thoracic and upper 3 lumbar segments
C-	The 2nd to 4th sacral segments
D-	The upper 4 thoracic segments of the spinal cord

22-	Sympathetic stimulation of the heart: (Important M3WAN - Core)
A-	Decreases all properties of cardiac muscle
B-	Increases contraction strength only, with no change in excitability or conductivity
C-	Increases all properties of cardiac muscle: contraction becomes stronger, and excitability and conductivity are increased
D-	Has no effect on the heart

23-	Sympathetic stimulation of the coronary vessels causes: (Important M3WAN - Core)
A-	Vasoconstriction, like most other vascular beds
B-	Vasodilation
C-	No change in coronary flow
D-	Complete coronary occlusion

24-	Sympathetic stimulation of the respiratory system causes: (M3WAN - Core)
A-	Inhibition of bronchial mucus secretion, bronchodilation, and vasoconstriction of pulmonary vessels
B-	Stimulation of bronchial mucus secretion and bronchoconstriction
C-	Bronchoconstriction with pulmonary vasodilation
D-	No effect on the airways or pulmonary vessels

25-	The splanchnic sympathetic division, which supplies abdominal and pelvic viscera, arises from the lateral horn cells of: (M3WAN - Core)
A-	The upper 2 thoracic segments
B-	The upper 4 thoracic segments
C-	The lower 6-7 thoracic and upper 3 lumbar segments of the spinal cord
D-	The 2nd to 4th sacral segments

26-	**Preganglionic fibers of the greater splanchnic nerve relay mostly in: (Important M3WAN - Core)**
A-	The superior cervical ganglion
B-	The lateral (paravertebral) ganglia only
C-	Terminal ganglia inside the target organ wall
D-	The collateral ganglia in the abdomen, except those supplying the adrenal medulla, which relay inside the gland itself

27-	**Stimulation of the greater splanchnic nerve causes vasoconstriction of the majority of arterioles in all of the following EXCEPT: (M3WAN - Hard)**		
A-	The stomach and small intestine	B-	The coronary vessels
C-	The kidneys	D-	The liver and pancreas

28-	**Stimulation of the greater splanchnic nerve on the gastrointestinal tract causes: (M3WAN - Core)**
A-	Relaxation of the walls of the stomach, small intestine, and proximal large intestine, with contraction of their sphincters, leading to food retention
B-	Contraction of the walls with relaxation of sphincters, promoting evacuation
C-	No effect on GI motility
D-	Increased GI secretions

29-	**Stimulation of the greater splanchnic nerve causes contraction of the smooth muscle in the capsule of the spleen, resulting in: (Important M3WAN - Core)**
A-	Sequestration of blood within the spleen
B-	Rupture of the splenic capsule
C-	No change in splenic blood volume
D-	Pouring of about 50 mL of stored concentrated blood into the bloodstream

30-	**Stimulation of the greater splanchnic nerve stimulates glycogenolysis in the liver, which: (Important M3WAN - Core)**		
A-	Decreases the blood glucose level	B-	Has no effect on blood glucose
C-	Increases the blood glucose level	D-	Increases only liver glycogen storage

31-	**Stimulation of the greater splanchnic nerve increases the metabolic rate (calorigenic effect) and also stimulates: (M3WAN - Core)**
A-	Lipogenesis in adipose tissue, decreasing free fatty acid levels
B-	Lipolysis in adipose tissue, increasing both blood glucose and free fatty acid levels
C-	Protein synthesis exclusively, with no effect on lipids
D-	Glycogen synthesis in muscle only

32-	**Stimulation of the greater splanchnic nerve to the adrenal medulla causes secretion of: (Important M3WAN - Core)**
A-	Catecholamines (adrenaline and noradrenaline)
B-	Cortisol and aldosterone
C-	Insulin and glucagon
D-	Thyroxine

33-	**The lesser splanchnic nerve arises from spinal segments: (M3WAN - Core)**
A-	T1 to T2, relaying in the superior cervical ganglion
B-	S2 to S4, relaying in terminal ganglia
C-	T6 to T9, relaying in the celiac ganglion
D-	L1 to L3, relaying in the inferior mesenteric ganglion

34-	**Stimulation of the lesser splanchnic nerve causes all of the following EXCEPT: (M3WAN - Hard)**
A-	Retention of stool by relaxation of the rectal wall and contraction of the internal anal sphincter
B-	Retention of urine by relaxation of the bladder wall and contraction of the internal ureteral sphincter
C-	Relaxation of the internal anal and urethral sphincters
D-	Ejaculation from the sex organs

35-	**Skeletal muscles receive sympathetic innervation from: (Important M3WAN - Core)**
A-	Both noradrenergic sympathetic fibers (mainly causing vasoconstriction) and cholinergic sympathetic fibers (causing strong vasodilation)
B-	Noradrenergic fibers only
C-	Cholinergic fibers only
D-	Purely parasympathetic fibers

36-	**During hemorrhage, sympathetic stimulation of skeletal muscle blood vessels mainly causes: (Important M3WAN - Core)**
A-	Vasodilation, which helps increase blood supply to muscles
B-	Vasoconstriction, which helps elevate arterial blood pressure
C-	No change in vascular tone
D-	Alternating vasoconstriction and vasodilation with no net effect

37-	**During muscular exercise, sympathetic stimulation of skeletal muscle blood vessels mainly causes: (Important M3WAN - Core)**
A-	Vasoconstriction, which decreases blood supply to muscles
B-	No change in vascular tone
C-	Complete vascular shutdown

D-	Vasodilation, which increases the blood supply to active muscles

38-	**The Orbeli phenomenon refers to the fact that sympathetic stimulation of skeletal muscles causes: (Important M3WAN - Core)**
A-	Weaker contraction and earlier fatigue
B-	No change in muscle contraction properties
C-	Better contraction, delayed fatigue, and early recovery after fatigue
D-	Immediate paralysis of the muscle

39-	**Preganglionic autonomic nerve fibers are classified as: (Important M3WAN - Core)**		
A-	Type: A (alpha) fibers	B-	Type: B fibers
C-	Type: A (delta) fibers	D-	Type: C fibers

40-	**The preganglionic sympathetic nerve fibers that innervate the eye originate from: (Important M3WAN - Core)**		
A-	Thoracic segments of the spinal cord	B-	The superior cervical ganglion
C-	Superior colliculi	D-	The ciliary ganglion

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	**C**	**B**	**A**	**C**	**D**	**B**	**A**	**D**	**C**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**A**	**D**	**C**	**A**	**B**	**D**	**C**	**B**	**A**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	**C**	**B**	**A**	**C**	**D**	**B**	**A**	**D**	**C**
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	**A**	**D**	**C**	**A**	**B**	**D**	**C**	**B**	**A**

ANS III (Parasympathetic nervous system)

Pre Exams & Formative

1-	Vagus nerve stimulation? (MID 31, FINAL 34)
A-	Reduces pancreatic and HCl secretion
B-	Increases HCl secretion and causes tachycardia
C-	Causes bradycardia and increases bile secretion
D-	All of the above are correct

2-	Which of these is nota neurotransmitter in the autonomic nervous system? (FINAL 31)						
A-	Acetylcholine	B-	Epinephrine	C-	Norepinephrine	D-	Muscarine

3-	Which type of receptors is located on sinoatrial (SA) nodal cells and mediates an increase in heart rate? (FINAL 31)						
A-	Alpha	B-	Beta	C-	Muscarinic	D-	Nicotinic

4-	Concerning cholinergic receptors? (FINAL 31, FINAL 32, 2ND TRIAL 32)
A-	The cholinergic receptors at the motor end plate (of skeletal muscle) are muscarinic
B-	The cholinergic receptors of sweat glands are nicotinic
C-	The cholinergic receptors of the adrenal medulla are muscarinic
D-	The cholinergic receptors of the autonomic ganglia are nicotinic

5-	Noradrenaline? (FINAL 31, FINAL 32)
A-	Increases intestinal motility
B-	Is released by postganglionic sympathetic nerve endings
C-	Isa potent vasodilator
D-	Is released from adrenal medulla only

6-	Atropine injection (drug which block muscarinic receptors) causes all of the following EXCEPT? (FINAL 31, FINAL 32, 2ND TRIAL 32)		
A-	Decreases the intestinal motility	B-	Increases secretion of sweat glands
C-	Bronchiolar dilatation	D-	Pupil dilatation

7-	Increases the activity of the parasympathetic nervous system causes? (FINAL 31)		
A-	Bronchoconstriction	B-	Increases the heart rate
C-	Dilatation of the pupil	D-	Decreases the salivary secretion

8-	Stimulation of oculomotor nerve causes? (FINAL 31)		
A-	Pupil dilatation	B-	Accommodation
C-	Increased lacrimation	D-	Widening of palpebral fissure

9-	Which of the following enzymes catalyzes the synthesis of Ach? (FINAL 32)		
A-	Acetyl-coenzymeA	B-	Acetylcholinesterase (AChE)
C-	Amino acid decarboxylase	D-	Choline acetyltransferase

10-	Vagus nerve? (FINAL 32)		
A-	increases intestinal motility	B-	acts as afferent fiber for defecation
C-	increases salivary secretion	D-	all of the above

11-	Stimulation of the parasympathetic neurons leads to an increase of? (FINAL 32)						
A-	GIT motility	B-	heart rate	C-	blood pressure	D-	Sweating

12-	Vagus nerve? (2ND TRIAL 32)		
A-	cut does not affect stomach	B-	acts as afferent fiber for defecation
C-	increases intestinal secretion	D-	decreases salivary secretion
E-	skeletal muscle		

13-	Which of the following is correct? (2ND TRIAL 32)
A-	Noradrenaline (NE) is secreted from the preganglionic sympathetic fibers
B-	Ach is inactivated by the enzyme acetyl cholinesterase
C-	Ach binds to muscarinic receptors only
D-	autonomic ganglia are located inside the CNS
E-	the receptors in the autonomic ganglia are muscarinic

14-	Stimulation of the pelvic nerve causes? (MID 33)
A-	Bronchodilation
B-	Contraction of the internal urethral & rectal (anal) sphincters
C-	V'D. of the blood vessels in the external genitalia
D-	Ejaculation

15-	Muscarinic receptors can be distinguished from nicotinic receptors because muscarinic receptors are: (FINAL 34)
A-	Stimulated by muscarine and blocked by epinephrine.
B-	Stimulated by muscarine and blocked by nicotine.
C-	Blocked by atropine and stimulated by nicotine.

D-	Blocked by atropine and stimulated by Ach

16-	**atropin cause all the following except (FINAL 34)**
A-	contract pupillary muscle

17-	**Generalized parasympathetic stimulation causes? (2ND TRIAL 35)**		
A-	Increase the heart rate	B-	Hypotension
C-	Pupillary dilatation	D-	Decrease the intestinal peristaltic movement

18-	**Facial nerve? (FINAL 36)**		
A-	Stimulate lacrimal secretion	B-	Increase sweat
C-	Vasoconstriction	D-	Pupil constriction

19-	**In fight or flight situation, heart rate increases by which receptor? (FINAL 36)**						
A-	Beta	B-	Alpha	C-	Nicotinic	D-	Muscarinic

20-	**Pelvic nerve supply all except (FINAL SNU 1)**						
A-	Ejaculation	B-	Erection	C-	defecation	D-	micturition

21-	**Vegas nerve has these effects except: (FINAL SNU 1)**
A-	coronary BVs vasoconstriction
B-	Increase gastric secretion
C-	decrease all properties of whole cardiac muscle

22-	**The receptors located on the cells of the Adrenal Medulla are: (FINAL SNU 1)**		
A-	Muscarinic receptors.	B-	Nicotinic receptors.
C-	Alpha-adrenergic receptors.	D-	Beta-adrenergic receptors.

23-	**The pelvic nerve stimulation (FINAL 37)**
A-	Contraction in wall and sphincter of exterior genitalia
B-	Vasoconstriction in the blood vessels of penile
C-	Vasodilation of the wall of penile and erection

24-	**Alpha receptor when stimulated? (FINAL 37)**		
A-	Relaxation in the sphincters of GIT	B-	Excitatory effect on heart
C-	Decrease gastrointestinal wall motility		

25-	**The chemical transmitter: (Formative 36)**
A-	at the neuromuscular junction and sweat gland is Ach.
B-	all of the above are correct
C-	in some central synapses is noradrenaline.
D-	in all autonomic ganglia is Ach.

26-	**Which of the following enzymes catalyzes the destruction of ACh? (Formative 36)**		
A-	Acetylcholinesterase (AChE)	B-	Choline acetyltransferase
C-	Amino acid decarboxylase	D-	Acetyl'coenzymeA

27-	**Which of the following responses is mediated by parasympathetic muscarinic receptors? (Formative 37)**
A-	Dilation of bronchiolar smooth muscle
B-	Constriction of gastrointestinal (GI) sphincters
C-	Miosis
D-	Ejaculation

28-	**A42-year-old man with myasthenia gravis notes increased muscle strength when he is treated with acetylcholinesterase inhibitor. The basis for his improvement is increased (Formative 37)**
A-	number of ACh receptors on the muscle end plates
B-	amount of acetylcholine (ACh) released from motor nerves
C-	amount of norepinephrine released from motor nerves
D-	levels of ACh at the muscle end plates

29-	**Which of the following constitutes 75% of the parasympathetic nerve fibers in the body? (2ND TRIAL 37)**
A-	Vagus nerve
B-	Oculomotor nerve
C-	Facial nerve
D-	Glossopharyngeal nerve

30-	**The vagus nerve has all of the following effects EXCEPT: (2ND TRIAL 37)**
A-	Decreased all properties of the whole cardiac muscle
B-	Bronchoconstriction
C-	Contraction of the stomach wall
D-	Increased hepatic blood flow

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
C	D	B	D	B	B	A	B	D	D
11	12	13	14	15	16	17	18	19	20
A	C	B	C	D	A	B	A	A	A
21	22	23	24	25	26	27	28	29	30
A	B	C	C	B	A	C	D	A	A

Department Qs

1-	**All the following structures are supplied by sympathetic system only EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Ventricles of the heart.	B-	Constrictor of pupillae muscle.
C-	Adrenal medulla.	D-	Skin.

2-	**Which of the following organs receive both sympathetic & parasympathetic: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Adrenal medulla.	B-	Sweat glands.
C-	Pilomotor muscles.	D-	Lungs.

3-	**Concerning the autonomic ganglia, all the following is true EXCEPT: (SNU Bank, Dr.Seham)**
A-	They acting as distributing centers.
B-	They are 3 types.
C-	Their chemical transmitter is acetylcholine.
D-	Their receptors are specifically blocked by atropine.

4-	**The parasympathetic nerves: (SNU Bank, Department Qs, Dr.Seham)**
A-	Are important in emergency conditions e'g' fight & flight reactions.
B-	Tend to have longer postgang. fibres than pregang. fibres.
C-	Are catabolic in their actions.
D-	Generally exert opposite effects to those of the symp. nerves.

5-	**Stimulation of the pelvic nerve causes : (SNU Bank, Department Qs, Dr.Seham)**
A-	Contraction of the bronchial muscle.
B-	Contraction of the internal urethral & rectal (anal) sphincters.
C-	V'D. of the blood vessels in the external genitalia.
D-	Stoppage of prostatic secretion.

6-	**Which of the following may be produced by parasymp. stimulation?: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Acceleration of the heart.	B-	Contraction of the urinary bladder wall.
C-	V'D. of skeletal muscle blood vessels.	D-	Dilatation of the pupil.

7-	**Stimulation of the pelvic nerve (sacral autonomic) produces: (SNU Bank, Department Qs, Dr.Seham)**
A-	Relaxation of the wall of rectum & contraction of the internal anal sphincter.
B-	V'C. of the blood vessels in the external genitalia causing erection of penis.
C-	Contraction of the urinary bladder wall & relaxation of the internal uretheral sphincter.
D-	Contraction of smooth muscles in the male sexual passages causing ejaculation.

8-	**Concerning the pelvic nerve, all the following is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	It is the nerve of erection (=nervus erigentes).
B-	It is motor to the urinary bladder & distal half of the large intestine.
C-	It is inhibitory to both the internal anal & urethral sphincters.
D-	It is motor to the wall of gall bladder.

9-	**Vagus parasympathetic nerve supplies all the following structures EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**						
A-	Gall bladder.	B-	Stomach.	C-	Rectum.	D-	Pancreas.

10-	**All of the following about sacral autonomic outflow is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**		
A-	It is the nerve of defecation.	B-	It is the nerve of ejaculation.
C-	It is the nerve of micturition.	D-	It relays in terminal ganglia.

11-	**The functions of parasymp. innervation to the salivary glands is: (SNU Bank, Department Qs, Dr.Seham)**		
A-	V'C. & secretion.	B-	V'D. & secretion.
C-	V'D. & inhibition of secretion.	D-	Secretion of scanty viscous saliva.

12-	**Facial nerve: (SNU Bank, Department Qs, Dr.Seham)**
A-	Stimulation increases the power of the lens of the eye.
B-	Supplies the lacrymal gland.
C-	Supplies the parotid salivary gland.
D-	Contains sympathetic postganglionic fibres.

13-	**Concerning the cranial parasympathetic outflow: (SNU Bank, Department Qs, Dr.Seham)**
A-	preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
B-	preganglionic outflow in the glossopharyngeal nerve arises from superior salivary nucleus.
C-	They relax the sphincter of Oddi.
D-	They cause bronchodilatation.

14-	**All of the following organs receive their parasympathetic innervation via the vagus nerves EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Gall bladder.	B-	Atria of the heart.
C-	Proximal half of large intestine.	D-	Rectum.

15-	**Salivary secretion increases on stimulation of: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Trigeminal nerve.	B-	Glossopharyngeal nerve.
C-	Vagus nerve.	D-	Hypoglossal nerve.

16-	**Vagal stimulation causes: (SNU Bank, Department Qs, Dr.Seham)**
A-	Tachycardia.
B-	Dilatation of the bronchi and bronchioles.
C-	Inhibition of plain muscles in stomach and small intestine.
D-	Increase in gastric and pancreatic secretion.

17-	**Stimulation of pelvic nerve causes: (SNU Bank, Department Qs, Dr.Seham)**
A-	Contraction of the internal urethral sphincter.
B-	Contraction of the internal anal sphincter.
C-	Contraction of bronchial muscle.
D-	Dilatation of the blood vessels of the external genitalia.

18-	**Motility of the intestine is increased by: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Adrenaline.	B-	Parasympathetic stimulation.
C-	Sympathetic blockers.	D-	Adrenal medulla secretions.

19-	**Acetylcholine action at the autonomic ganglia is: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Muscarinic like action.	B-	Nicotine like action.
C-	Blocked by atropine.	D-	Blocked by muscarine.

20-	**Nicotinic receptors: (SNU Bank, Department Qs, Dr.Seham)**
A-	Accept nicotine released by nerve endings near the receptor.
B-	Are not sensitive to accetylcholine.

C-	Are blocked by atropine.
D-	Are found in sympathetic ganglia.

21-	**About acetylcholine, all the following is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	It is synthesized & released by some symp. postgang. nerve fibres.
B-	It is the chemical transmitter released by the pregang. nerve fibres.
C-	It is broken down enzymatically by the cholinestrase enzyme.
D-	It makes reading difficult because of relaxation of the ciliary muscle.

22-	**An injection of atropine causes all the following effects EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**		
A-	An increase in the heart rate.	B-	Pupillo-dilatation.
C-	Difficult micturition.	D-	Constriction of the bronchi.

23-	**About acetylcholine, all the following is true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	Its synthesis is catalyzed by the choline acetyltransferase enzyme.
B-	Its muscarinic action can be blocked by atropine.
C-	Its nicotinic actions can be blocked by large doses of nicotine.
D-	It is not secreted by any symp. postgang. nerve fibres.

24-	**The nicotinic action of acetylcholine: (SNU Bank, Department Qs, Dr.Seham)**
A-	Is blocked by atropine.
B-	Is excitatory to both intestinal movements & secretion.
C-	Is stimulatory to the autonomic ganglia.
D-	Is not affected by curare.

25-	**Acetylcholine is secreted at all the following sites EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	Postgang. sympathetic nerve terminals at skeletal muscle.
B-	Postgang. parasymp. nerve terminals.
C-	Pregang. symp. & parasymp. nerve terminals.
D-	Postgang. symp. noradrenergic nerve terminals.

26-	**The nicotinic-like action is present in: (SNU Bank, Department Qs, Dr.Seham)**
A-	All preganglionic nerve endings.
B-	The symp. postgang. supply to the blood vessels in skeletal muscles.
C-	The parasymp. postgang. supply to the heart.
D-	The symp. postgang. supply to the eccrine sweat glands.

27-	About Prostigmine (Neostigmine), all are true EXCEPT: (SNU Bank, Department Qs, Dr.Seham)		
A-	Acts specifically at motor end plate.	B-	Is used in treatment of myasthenia gravis.
C-	Inhibits cholinesterase enzyme.	D-	Is an irreversible anticholinesterase.

28-	The antidote of eserine is: (SNU Bank, Department Qs, Dr.Seham)						
A-	Amphetamine.	B-	Noradrenaline.	C-	Ergotoxine.	D-	Atropine.

29-	Atropine injection causes, all the following EXCEPT: (SNU Bank, Department Qs, Dr.Seham)		
A-	Pupillodilatation.	B-	Increased heart rate.
C-	Relaxation of skeletal muscles.	D-	Diminished salivary secretion.

30-	Muscarine-like action of acetyl choline is present in all the following EXCEPT: (SNU Bank, Department Qs, Dr.Seham)
A-	Sympathetic postganglionic supplying the skeletal muscle blood vessels.
B-	Parasympathetic postaganglionic nerve endings to the heart.
C-	Sympathetic postganglionic supplying the eccrine sweat glands.
D-	Somatic neuromuscular junction.

31-	Which of the following is used in treatment of myasthenia gravis: (SNU Bank, Department Qs, Dr.Seham)						
A-	Prostigmine.	B-	Eserine.	C-	Atropine	D-	Curare.

32-	Curare: (SNU Bank, Department Qs, Dr.Seham)
A-	Blocks the neuromuscular transmission.
B-	Completes with acetyl choline at autonomic ganglia..
C-	Prevents the release of acetyl choline at motor end plate.
D-	Causes relaxation of smooth muscles.

33-	Atropine blocks the action of acetyl choline at the following sites EXCEPT: (SNU Bank, Department Qs, Dr.Seham)		
A-	Heart.	B-	Adrenal medulla.
C-	Sweat glands.	D-	Constrictor pupillae muscle.

34-	About acetyl choline, all the following are true EXCEPT: (SNU Bank, Department Qs)
A-	Is synthesized in the adrenal medulla.
B-	Released by all postganglionic parasympathetic nerves.
C-	Inactivated by cholinesterase enzyme.
D-	Released by the sympathetic cholinergic postganglionic nerve fibres.

35-	Atropine is given before surgical operations to: (SNU Bank, Department Qs, Dr.Seham)		
A-	decrease heart rate.	B-	Prevent bronchial secretion.
C-	Relax skeletal muscle.	D-	Induce bronchconstriction.

36-	Atropine: (SNU Bank, Department Qs, Dr.Seham)		
A-	Increases the tone & motility of GIT.	B-	Decreases gastric acid secretion.
C-	Stimulates sweat secretion.	D-	Stimulates bronchial secretion.

37-	Administration of physostigmine would be expected to produce all the following EXCEPT: (SNU Bank, Department Qs)
A-	Increase the secretion of gastric juice.
B-	Increase the rate of epinephrine secretion from adrenal medulla.
C-	Cause hypotension & bradycardia.
D-	Produce lacrymation & bronchodilatation.

38-	Nicotine in small doses: (SNU Bank, Department Qs)
A-	Acts directly on intestinal smooth muscle, causing it to contract.
B-	Decreases the release of acetyl choline in sympathetic ganglia.
C-	Increases the rate of discharge of postganglionic parasympathetic neurones.
D-	Decreases the rate of discharge of postganglionic sympathetic neurones.

39-	Atropine causes: (SNU Bank, Department Qs)
A-	Contraction of ciliary muscle.
B-	Dryness of mouth & skin.
C-	Bradycardia during rest.
D-	Bronchodilatation & increases bronchial secretion.

40-	Cholinergic fibres include all the following EXCEPT: (SNU Bank, Department Qs, Dr.Seham)
A-	Sympathetic & parasympathetic preganglionic nerve fibres.
B-	Somatic motor nerve fibres.
C-	Sympathetic nerve fibres to adrenal medulla.
D-	Postganglionic nerve fibres to dilator pupilla muscle.

41-	Acetyl choline: (SNU Bank, Department Qs, Dr.Seham)
A-	Release is blocked by curare at motor end plate.
B-	Synthesis is catalyzed by cholinesterase enzyme.
C-	Action at motor end plate is blocked by atropine.

D-	Is the only neurotransmitter secreted by parasympathetic neurones.

42-	**Stimulation of alpha-adrenergic receptors produces all the following EXCEPT: (SNU Bank, Department Qs)**		
A-	Contraction of dilator pupillae muscle.	B-	Dilatation of skeletal muscle blood vessels.
C-	Constriction of cutaneous blood vessels.	D-	Ejaculation of semen.

43-	**Concerning adrenergic receptors, all the following are true EXCEPT: (SNU Bank, Department Qs)**
A-	Stimulation of 1 adrenoceptors increases the force & rate of cardiac contraction.
B-	Activation of 1 & 2 adrenoceptors results in activation of adenyl cyclase.
C-	Stimulation of alpha adrenoceptors causes bronchodilatation.
D-	Stimulation of 1 receptors causes renin secretion & lipolysis.

44-	**Catecholamines include all the following EXCEPT: (SNU Bank, Department Qs)**						
A-	Epinephrine.	B-	Norepinephrine.	C-	Dopamine.	D-	Ephedrine.

45-	**The following structures respond to beta adrenergic receptors stimulation EXCEPT: (SNU Bank, Department Qs)**		
A-	Sinoatrial node.	B-	Skeletal muscle blood vessels.
C-	Dilator pupillae muscle.	D-	Bronchial smooth muscles

46-	**All the following drugs would be expected to increase sympathetic discharge or mimic the effects of increased sympathetic discharge EXCEPT: (SNU Bank, Department Qs)**						
A-	Norepinephrine.	B-	Neostigmine.	C-	Amphetamine.	D-	Tyramine.

47-	**Noradrenaline: (SNU Bank, Department Qs)**
A-	Increases intestinal motility.
B-	Is the neurotransmitter in the alpha-adrenergic receptors.
C-	Is released by both pre- & postganglionic sympathetic nerve endings.
D-	Isa potent vasodilator.

48-	**Administration of beta blockers produce all the following EXCPT: (SNU Bank, Department Qs)**
A-	May cause bronchoconstriction.
B-	Produces bradycardia & lowers the arterial blood pressure.
C-	Decreases the secretion of renin from the kidney.
D-	Leads to bronchodilatation.

49-	**Concerning Epinephrine, all the following are true EXCEPT: (SNU Bank, Department Qs)**
A-	Is synthesized in adrenal medulla by methylation of norepinephrine.
B-	Is the chemical transmitter of adrenergic postganglionic sympathetic fibres.
C-	Is less potent than norepinephrine as a pressor agent.
D-	Helps in treatment of bronchial asthma.

50-	**Concerning norepinephrine, all are true EXCEPT: (SNU Bank, Department Qs)**
A-	It contributes to 20% of the suprarenal medulla secretion.
B-	It is secreted under stressful conditions.
C-	It is removed from tissues by active reuptake into sympathetic nerve endings.
D-	It is the chemical transmitter at most sympathetic preganglionic.

51-	**As regards autonomic receptors, all are true EXCEPT: (SNU Bank, Department Qs)**
A-	Sweat glands innervated by sympathetic nerves have receptors that can be blocked by atropine.
B-	Propranolol blocks the action of noradrenaline on alpha receptors.
C-	Nicotinic receptors are blocked following a large dose of nicotine.
D-	Alpha adrenoceptors are blocked by phentolamine or ergotamine.

52-	**Concerning adrenergic receptors, all are true EXCEPT: (SNU Bank, Department Qs)**
A-	Are receptors, which responds norepinephrine secreted by nerve ending & by adrenal medulla.
B-	Are classified into & adrenoceptors.
C-	Trigger the release of intracellular mediator (2nd messenger) when activated.
D-	When stimulated always produces excitatory effects on the effector organs.

53-	**A drug which blocks beta adrenoceptors is likely to cause: (SNU Bank, Department Qs)**
A-	Dilatation of bronchi.
B-	Decrease heart rate.
C-	Ciliary muscle relaxation for far vision.
D-	Reduces blood sugar level due to stimulation of insulin secretion.

54-	**All of the following statements are true EXCEPT: (SNU Bank, Department Qs)**
A-	Nicotinic receptors of acetyl choline can be blocked by large dose of nicotine.
B-	Muscarinic receptors of acetyl choline are blocked by atropine.
C-	Alpha adrenergic receptors are blocked by phentolamine & ergotamine.
D-	Beta adrenergic receptors are blocked by guanethidine.

55-	**Epinephrine is : (SNU Bank, Department Qs)**
A-	Synthesized in the adrenal medulla by methylation of norepinephrine.
B-	The chemical transmitter of most symp. postgang. nerve fibres.
C-	More powerful as a pressor agent than norepinephrine.
D-	Hydrolyzed by the cholinestrase enzyme.

56-	**About β-adrenergic receptors which is false: (SNU Bank, Department Qs)**
A-	Are not stimulated by epinephrine.
B-	When stimulated, they reduce the gastrointestinal (GI) motility.
C-	Are blocked by propranolol.
D-	When stimulated, they accelerate the heart.

57-	**Removal of norepinephrine at postgang. symp. fibres occurs by: (SNU Bank, Department Qs)**
A-	Reuptake back into the terminal symp. nerve endings.
B-	Transformation into epinephrine.
C-	The action of transaminase enzyme.
D-	Ademethylation action.

58-	**Epinephrine produces all the following effects EXCEPT: (SNU Bank, Department Qs)**		
A-	Cutaneous V'C.	B-	Brobchoconstriction.
C-	Coronary V'D.	D-	Glycogenolysis.

59-	**The α-adrenergic receptors mediate all the following effects EXCEPT: (SNU Bank, Department Qs)**		
A-	Vasoconstriction.	B-	Pupil dilatation.
C-	Spleen contraction.	D-	Increased cardiac activity.

60-	**All the following drugs augment (increase) the sympathetic activity EXCEPT: (SNU Bank, Department Qs)**						
A-	Reserpine.	B-	Ephedrine.	C-	Amphetamine.	D-	Tyramine.

61-	**Administration of noradrenaline toa person whose symp. System has been surgically removed reacts by developing: (SNU Bank, Department Qs)**
A-	Hypoglycaemia.
B-	Hypotension.
C-	Pupil constriction.
D-	More marked sympathetic response than before sympathectomy.

62-	Which is NOT a neurotransmitter in ANS? (SNU Bank)						
A-	Acetylcholine	B-	Epinephrine	C-	Norepinephrine	D-	Muscarine

63-	Which receptor on SA node increases heart rate? (SNU Bank)						
A-	Alpha	B-	Beta	C-	Muscarinic	D-	Nicotinic

64-	Concerning cholinergic receptors? (SNU Bank)		
A-	Motor end plate receptors are muscarinic	B-	Sweat gland receptors are nicotinic
C-	Adrenal medulla receptors are muscarinic	D-	Autonomic ganglia receptors are nicotinic

65-	Noradrenaline? (SNU Bank)
A-	Increases intestinal motility
B-	Is released by postganglionic sympathetic nerve endings
C-	Isa potent vasodilator
D-	Is released from adrenal medulla only

66-	Atropine (muscarinic blocker) causes all EXCEPT? (SNU Bank)		
A-	Decreases intestinal motility	B-	Increases secretion of sweat glands
C-	Bronchiolar dilatation	D-	Pupil dilatation

67-	Atropine injection (blocks muscarinic receptors) causes all EXCEPT? (SNU Bank)		
A-	Decreases intestinal motility	B-	Bronchiolar dilatation
C-	Increases secretion of sweat glands	D-	Pupil dilatation
E-	Skeletal muscle		

68-	Which fibers release Norepinephrine? (SNU Bank)
A-	Preganglionic fibers to adrenal medulla
B-	Postganglionic fibers causing bradycardia
C-	Postganglionic fibers causing iris constriction
D-	Postganglionic fibers causing arteriolar constriction

69-	Increased parasympathetic activity causes? (SNU Bank)		
A-	Bronchoconstriction	B-	Increases heart rate
C-	Dilatation of pupil	D-	Decreases salivary secretion

70-	Stimulation of oculomotor nerve causes? (SNU Bank)		
A-	Pupil dilatation	B-	Accommodation
C-	Increased lacrimation	D-	Widening of palpebral fissure

71-	Which enzyme catalyzes synthesis of ACh? (SNU Bank)		
A-	Acetyl-coenzymeA	B-	Acetylcholinesterase (AChE)
C-	Amino acid decarboxylase	D-	Choline acetyltransferase

72-	Vagus nerve? (SNU Bank)		
A-	Increases intestinal motility	B-	Acts as afferent fiber for defecation
C-	Increases salivary secretion	D-	All of the above

73-	Parasympathetic stimulation increases? (SNU Bank)						
A-	GIT motility	B-	Heart rate	C-	Blood pressure	D-	Sweating

74-	Generalized parasympathetic stimulation causes? (SNU Bank)		
A-	Increase the heart rate	B-	Hypotension
C-	Pupillary dilatation	D-	Decrease the intestinal peristaltic movement

75-	Facial nerve? (SNU Bank)		
A-	Stimulate lacrimal secretion	B-	Increase sweat
C-	Vasoconstriction	D-	Pupil constriction

76-	In fight or flight situation, heart rate increases by which receptor? (SNU Bank)						
A-	Beta	B-	Alpha	C-	Nicotinic	D-	Muscarinic

77-	Concerning the autonomic ganglia, all the following is true EXCEPT: (SNU Bank)
A-	They acting as distributing centers
B-	They are 3 types
C-	Their chemical transmitter is acetylcholine
D-	Their receptors are specifically blocked by atropine

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	D	D	D	C	B	C	D	C	B
11	12	13	14	15	16	17	18	19	20
B	B	C	D	B	D	D	B	B	D
21	22	23	24	25	26	27	28	29	30
D	D	D	C	D	A	D	D	C	D
31	32	33	34	35	36	37	38	39	40
A	A	B	A	B	B	D	C	B	D

41	42	43	44	45	46	47	48	49	50
D	B	C	D	C	B	B	D	B	D
51	52	53	54	55	56	57	58	59	60
B	D	B	D	A	A	A	B	D	A
61	62	63	64	65	66	67	68	69	70
D	D	B	D	B	B	C	D	A	B
71	72	73	74	75	76	77	78	79	80
D	A	A	B	A	A	D	-	-	-

M3WAN Bank

1-	**The parasympathetic nervous system is primarily concerned with body activities during: (M3WAN - Core)**
A-	Exercise, Excitement, Emergency, and Embarrassment (the 'E' division)
B-	Fight or flight reactions only
C-	Growth and development exclusively
D-	Digestion, Defecation, and Diuresis (the 'D' division, 'rest or digest')

2-	**The parasympathetic nervous system is distributed through: (M3WAN - Core)**
A-	A cranial division and a sacral outflow
B-	A thoracic division and a lumbar outflow
C-	Asingle cervical division only
D-	Four divisions identical to the sympathetic system

3-	**The parasympathetic supply to the eye arises from the oculomotor nucleus and relays in the: (M3WAN - Core)**		
A-	Superior cervical ganglion	B-	Ciliary ganglion
C-	Otic ganglion	D-	Sphenopalatine ganglion

4-	**Parasympathetic stimulation via the oculomotor nerve (III) causes contraction of the constrictor pupillae muscle, leading to: (Important M3WAN - Core)**		
A-	Mydriasis	B-	Exophthalmos
C-	Miosis	D-	Widening of the palpebral fissure

5-	**Parasympathetic stimulation via the oculomotor nerve (III) causes contraction of the ciliary muscle, which mediates: (M3WAN - Core)**		
A-	Far vision	B-	Dark adaptation
C-	Pupillary dilation	D-	Near vision (accommodation)

6-	**The facial nerve (VII) carries parasympathetic preganglionic fibers arising from the superior salivary nucleus and relays in the: (M3WAN - Core)**		
A-	Otic ganglion only	B-	Ciliary ganglion
C-	Sphenopalatine and submandibular ganglia	D-	Superior cervical ganglion

7-	**Postganglionic parasympathetic fibers of the facial nerve (VII) supply: (M3WAN - Core)**
A-	The parotid salivary gland
B-	The lacrimal gland only
C-	The sublingual and submandibular salivary glands
D-	The pancreas

8-	**In addition to its secretomotor function, the facial nerve (VII) carries taste sensation from: (M3WAN - Core)**		
A-	The posterior third of the tongue	B-	The entire tongue equally
C-	The anterior two-thirds of the tongue	D-	The soft palate only

9-	**The glossopharyngeal nerve (IX) carries parasympathetic preganglionic fibers arising from the inferior salivary nucleus and relays in the: (M3WAN - Core)**		
A-	Ciliary ganglion	B-	Sphenopalatine ganglion
C-	Submandibular ganglion	D-	Otic ganglion

10-	**Postganglionic parasympathetic fibers of the glossopharyngeal nerve (IX) supply: (M3WAN - Core)**		
A-	The submandibular salivary gland	B-	The sublingual salivary gland
C-	The parotid salivary gland	D-	The lacrimal gland

11-	**The glossopharyngeal nerve (IX) carries taste sensation from: (M3WAN - Core)**		
A-	The anterior two-thirds of the tongue	B-	The posterior third of the tongue
C-	The entire tongue	D-	The epiglottis only

12-	**The net result of parasympathetic stimulation of the salivary glands (via facial and glossopharyngeal nerves) is: (Important M3WAN - Core)**
A-	Trophic secretion: little in amount, viscid, and rich in enzymes
B-	Complete cessation of salivary flow

C-	True secretion: profuse, watery, and little in enzymes
D-	Secretion identical to sympathetic trophic secretion

13-	**The vagus (10th cranial) nerve constitutes approximately what proportion of the parasympathetic fibers in the body? (Important M3WAN - Core)**						
A-	25%	B-	50%	C-	90%	D-	75%

14-	**Preganglionic fibers of the vagus nerve arise from the dorsal vagus nucleus in the medulla oblongata and relay in: (M3WAN - Core)**		
A-	The superior cervical ganglion	B-	Collateral ganglia
C-	Lateral (paravertebral) ganglia	D-	Terminal ganglia

15-	**Vagal stimulation of the heart: (Important M3WAN - Core)**
A-	Increases heart rate and contractility
B-	Has no effect on the atria or ventricles
C-	Inhibits all properties of cardiac muscle, slowing heart rate and decreasing contractility, excitability, and conductivity
D-	Selectively increases ventricular excitability only

16-	**Vagal inhibitory effects on the heart are limited to: (Important M3WAN - Core)**
A-	The ventricles only
B-	The atria only, since the vagi do not supply the ventricles
C-	Both atria and ventricles equally
D-	Neither the atria nor the ventricles

17-	**The phenomenon in which the vagus nerve does not supply the ventricles, so that vagal effects on the heart are confined to the atria, is called: (Important M3WAN - Core)**		
A-	Vagus escape	B-	Orbeli phenomenon
C-	Vagal blockade	D-	Cardiac accommodation

18-	**Vagal stimulation of the lungs causes: (Important M3WAN - Core)**
A-	Bronchodilation and vasoconstriction of pulmonary vessels
B-	Bronchoconstriction, mucus secretion from the bronchial glands, and vasodilation of pulmonary vessels
C-	No effect on the airways
D-	Inhibition of all bronchial gland secretion

19-	**Vagal stimulation of the gastrointestinal tract causes contraction of the walls (esophagus, stomach, small intestine, and proximal large intestine) and relaxation of the corresponding sphincters, thereby promoting: (M3WAN - Core)**
A-	Food retention and reduced GI secretion
B-	Deglutition (swallowing) and evacuation of food from the GIT
C-	Complete cessation of GI motility
D-	Sphincter contraction and food retention

20-	**Vagal stimulation of the gallbladder causes: (M3WAN - Core)**
A-	Relaxation of its wall and contraction of the sphincter of Oddi, preventing evacuation
B-	No effect on the gallbladder
C-	Contraction of its wall and relaxation of the sphincter of Oddi, helping evacuation
D-	Contraction of the sphincter of Oddi only

21-	**Vagal stimulation causes secretion from all of the following EXCEPT: (M3WAN - Hard)**
A-	The stomach (gastric juice)
B-	The pancreas (pancreatic juice, insulin, and glucagon)
C-	The adrenal medulla (catecholamines)
D-	The duodenum (mucus from Brunner's glands) and the liver (bile)

22-	**Vagal stimulation of the liver also produces: (M3WAN - Core)**
A-	Vasoconstriction, decreasing hepatic blood flow
B-	No vascular effect
C-	Vasodilation, increasing hepatic blood flow
D-	Cessation of bile secretion

23-	**The sacral parasympathetic outflow originates from: (Important M3WAN - Core)**
A-	The 2nd, 3rd, and 4th sacral segments of the spinal cord
B-	The upper 3 lumbar segments
C-	All thoracic segments
D-	Cranial nerve nuclei only

24-	**Preganglionic fibers of the sacral parasympathetic outflow pass through the sympathetic chain without relay and unite to form the: (Important M3WAN - Core)**		
A-	Greater splanchnic nerve	B-	Vagus nerve
C-	Pelvic nerve	D-	Lesser splanchnic nerve

25-	**The pelvic nerve relays in the: (M3WAN - Core)**		
A-	Terminal (hypogastric) ganglia	B-	Superior cervical ganglion
C-	Celiac ganglion	D-	Ciliary ganglion

26-	**Parasympathetic (pelvic nerve) stimulation of the urinary bladder causes: (Important M3WAN - Core)**
A-	Relaxation of the detrusor muscle and contraction of the internal urethral sphincter, causing urine retention
B-	Contraction of the detrusor muscle and relaxation of the internal urethral sphincter, resulting in micturition
C-	No effect on bladder function
D-	Contraction of both the detrusor muscle and the internal urethral sphincter

27-	**Parasympathetic (pelvic nerve) stimulation of the rectum and distal descending colon causes contraction of the walls and relaxation of the internal anal sphincter, resulting in: (M3WAN - Core)**		
A-	Constipation	B-	Defecation
C-	No effect on bowel movement	D-	Retention of stool

28-	**Parasympathetic (pelvic nerve) stimulation of the seminal vesicles and prostate causes: (M3WAN - Core)**		
A-	Secretion from these glands	B-	Contraction leading to ejaculation
C-	Shrinkage of these glands	D-	No effect on these glands

29-	**Parasympathetic (pelvic nerve) stimulation produces vasodilation of the erectile tissues (penis and clitoris), resulting in: (Important M3WAN - Core)**		
A-	Ejaculation	B-	Shrinkage of erectile tissue
C-	Erection	D-	No vascular change

30-	**The relationship between the sympathetic and parasympathetic divisions in organs such as the heart, lungs, and pelvis is usually: (M3WAN - Core)**		
A-	Synergistic	B-	Antagonistic
C-	Unrelated	D-	Identical in direction

31-	**Regarding salivary secretion, the relationship between sympathetic and parasympathetic stimulation is: (Important M3WAN - Core)**
A-	Antagonistic, since only one division can act at a time
B-	Synergistic: sympathetic produces trophic secretion while parasympathetic produces true secretion
C-	Purely sympathetic, with no parasympathetic contribution
D-	Purely parasympathetic, with no sympathetic contribution

32-	**Regarding the sexual act, the relationship between sympathetic and parasympathetic stimulation is synergistic: (Important M3WAN - Core)**
A-	Parasympathetic causes ejaculation while sympathetic causes erection
B-	Both divisions cause identical effects
C-	Sympathetic causes secretion while parasympathetic causes ejaculation
D-	Parasympathetic causes erection and glandular secretion, while sympathetic causes ejaculation and shrinkage

33-	**Structures supplied by the sympathetic division alone include all of the following EXCEPT: (Important M3WAN - Hard)**		
A-	The dilator pupillae muscle	B-	The constrictor pupillae muscle
C-	The ventricles of the heart	D-	The spleen and the adrenal medulla

34-	**The structure supplied by the parasympathetic division only, with no sympathetic supply, is: (Important M3WAN - Core)**		
A-	The dilator pupillae muscle	B-	The constrictor pupillae muscle
C-	The ventricles of the heart	D-	Most blood vessels

35-	**The two main chemical transmitters in the autonomic nervous system are: (M3WAN - Core)**		
A-	Dopamine and serotonin	B-	GABA and glutamate
C-	Acetylcholine and noradrenaline	D-	Histamine and glycine

36-	**Synthesis of acetylcholine from acetyl CoA and choline is catalyzed by the enzyme: (Important M3WAN - Core)**		
A-	Acetylcholinesterase	B-	Monoamine oxidase
C-	Catechol-O-methyltransferase	D-	Choline acetyltransferase

37-	**After synthesis, acetylcholine is stored inside the synaptic knob within: (M3WAN - Core)**		
A-	Vesicles	B-	The nucleus
C-	Mitochondria	D-	The Golgi apparatus

38-	**Release of acetylcholine from the presynaptic membrane occurs by exocytosis, triggered by: (M3WAN - Core)**		
A-	Opening of Na+ channels	B-	Opening of Ca2+ channels
C-	Closure of K+ channels	D-	Activation of the Na+/K+ pump

39-	The main enzyme responsible for degrading acetylcholine into choline and acetate is: (Important M3WAN - Core)		
A-	Monoamine oxidase	B-	Choline acetyltransferase
C-	Catechol-O-methyltransferase	D-	Acetylcholinesterase

40-	After acetylcholine breakdown, choline is: (M3WAN - Core)
A-	Excreted unchanged in urine
B-	Reuptaken by the presynaptic neuron and recycled for resynthesis of acetylcholine
C-	Converted directly into noradrenaline
D-	Permanently degraded with no further use

41-	True cholinesterase is characterized by being: (Important M3WAN - Core)
A-	Non-specific with rapid turnover, present in liver, spleen, and plasma
B-	Present only in the adrenal medulla
C-	Specific with slow turnover, present in nervous tissue and red blood cells
D-	Identical in distribution to pseudocholinesterase

42-	Pseudocholinesterase is characterized by being: (Important M3WAN - Core)
A-	Specific with slow turnover, present in nervous tissue and red blood cells
B-	Non-specific with rapid turnover, present in the liver, spleen, and plasma
C-	Present exclusively at the neuromuscular junction
D-	Identical in distribution to true cholinesterase

43-	Central cholinergic fibers, which release acetylcholine, are found at the endings of: (Important M3WAN - Core)
A-	All postganglionic parasympathetic fibers only
B-	Sympathetic postganglionic fibers to sweat glands only
C-	All preganglionic autonomic neurons (sympathetic and parasympathetic) and at neuromuscular junctions
D-	Beta-adrenergic receptor sites only

44-	Peripheral cholinergic fibers include all parasympathetic postganglionic nerve endings, together with sympathetic postganglionic fibers that supply: (Important M3WAN - Core)
A-	The heart and coronary vessels
B-	The adrenal medulla
C-	The dilator pupillae muscle
D-	Eccrine sweat glands and skeletal muscle blood vessels

45-	**Muscarinic cholinergic receptors are named after muscarine and are found mainly at: (Important M3WAN - Core)**		
A-	Peripheral cholinergic sites	B-	The neuromuscular junction exclusively
C-	All autonomic ganglia exclusively	D-	Central synapses only

46-	**Nicotinic cholinergic receptors are named after nicotine and are found mainly at: (Important M3WAN - Core)**
A-	Peripheral parasympathetic postganglionic endings only
B-	Beta-adrenergic sites
C-	Central cholinergic sites (autonomic ganglia and neuromuscular junctions)
D-	Alpha-adrenergic sites

47-	**Regarding the neurotransmitter released by sympathetic postganglionic neurons: (Important M3WAN - Core)**
A-	All sympathetic postganglionic neurons are cholinergic
B-	Most liberate noradrenaline, but some (e'g', to sweat glands and skeletal muscle blood vessels) are cholinergic
C-	All are exclusively dopaminergic
D-	None release any neurotransmitter, acting purely electrically

48-	**The synthesis pathway of catecholamines proceeds in the order: (Important M3WAN - Core)**
A-	Dopamine → DOPA → tyrosine → phenylalanine → noradrenaline → adrenaline
B-	Tyrosine → phenylalanine → DOPA → dopamine → adrenaline → noradrenaline
C-	Phenylalanine → tyrosine → DOPA → dopamine → noradrenaline → adrenaline
D-	Noradrenaline → adrenaline → dopamine → DOPA → tyrosine → phenylalanine

49-	**Release of noradrenaline from postganglionic sympathetic nerve terminals occurs by exocytosis, triggered by: (M3WAN - Core)**
A-	Ca2+ entry into the nerve terminal following arrival of an action potential
B-	Na+ entry into the nerve terminal
C-	K+ efflux from the nerve terminal
D-	Direct activation of beta receptors

50-	**Release of adrenaline and noradrenaline from the adrenal medulla is triggered by: (Important M3WAN - Core)**
A-	Direct sympathetic postganglionic innervation releasing noradrenaline onto the gland
B-	Circulating cortisol

C-	Acetylcholine released after stimulation of preganglionic sympathetic fibers in the greater splanchnic nerve
D-	Parasympathetic vagal stimulation

51-	**Inactivation of catecholamines occurs by all of the following mechanisms EXCEPT: (M3WAN - Hard)**
A-	Active re-uptake into the vesicles of the nerve endings that released them
B-	Oxidation by monoamine oxidase (MAO)
C-	Methylation by catechol-O-methyltransferase (COMT)
D-	Degradation by acetylcholinesterase

52-	**Alpha adrenergic receptors generally produce: (Important M3WAN - Core)**
A-	Inhibitory effects everywhere without exception
B-	Excitatory effects, such as vasoconstriction of blood vessels and contraction of the dilator pupillae muscle
C-	No physiological effect
D-	Effects identical to muscarinic receptors

53-	**Alpha-2 adrenergic receptors are notable because, unlike most alpha receptor effects, they particularly produce inhibitory effects in: (Important M3WAN - Hard)**
A-	The heart and coronary vessels
B-	Skeletal muscle blood vessels
C-	The stomach and intestinal walls, and the islets of Langerhans
D-	The dilator pupillae muscle

54-	**Beta-1 adrenergic receptors produce mainly excitatory effects and are located mainly in: (Important M3WAN - Core)**		
A-	The bronchial smooth muscle	B-	The heart
C-	The urinary bladder wall	D-	The spleen capsule

55-	**Beta-2 adrenergic receptors generally produce inhibitory effects, such as vasodilation and relaxation of the bronchial muscle, ciliary muscle, and the walls of the stomach, intestine, and urinary bladder; however, in certain sites (liver, heart, and islets of Langerhans) they produce: (Important M3WAN - Hard)**		
A-	No effect at all	B-	Excitatory effects
C-	Effects identical to alpha-2 receptors	D-	Only muscarinic-type effects

56-	**Beta-3 adrenergic receptors are found mainly in: (M3WAN - Hard)**		
A-	Adipose tissue	B-	The bronchial smooth muscle
C-	The urinary bladder wall	D-	The spleen capsule

57-	The primary higher control center for the autonomic nervous system, integrating information to regulate homeostasis through both the autonomic and endocrine systems, is the: (Important M3WAN - Core)						
A-	Cerebellum	B-	Hypothalamus	C-	Occipital cortex	D-	Corpus callosum

58-	Higher control centers of the autonomic nervous system include all of the following EXCEPT: (M3WAN - Hard)		
A-	The hypothalamus	B-	The limbic system
C-	The brainstem	D-	The occipital lobe

59-	Compared with noradrenaline, adrenaline: (Important M3WAN - Core)
A-	Binds primarily to alpha receptors with little effect on beta receptors
B-	Is produced mostly at sympathetic nerve endings rather than the adrenal medulla
C-	Binds to both alpha and beta adrenergic receptors more equally, producing wide-ranging effects
D-	Has no effect on the cardiovascular system

60-	Regarding the effects of adrenaline on the airways compared with noradrenaline: (M3WAN - Core)
A-	Adrenaline causes bronchodilation, while noradrenaline has little to no effect on the airways at therapeutic doses
B-	Both cause equally strong bronchodilation
C-	Adrenaline causes bronchoconstriction, while noradrenaline causes bronchodilation
D-	Neither hormone has any effect on airway diameter

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	A	B	C	D	C	C	C	D	C
11	12	13	14	15	16	17	18	19	20
B	C	D	D	C	B	A	B	B	C
21	22	23	24	25	26	27	28	29	30
C	C	A	C	A	B	B	A	C	B
31	32	33	34	35	36	37	38	39	40
B	D	B	B	C	D	A	B	D	B
41	42	43	44	45	46	47	48	49	50
C	B	C	D	A	C	B	C	A	C
51	52	53	54	55	56	57	58	59	60
D	B	C	B	B	A	B	D	C	A

REGULATION OF BODY TEMPRATURE

Pre Exams & Formative

1-	**On exposure to cold? (FINAL 31)**
A-	The muscle tone increases and shivering may occur
B-	There is excessive secretion of catecholamines
C-	The sympathetic is activated causing vasoconstriction & piloerection
D-	All the above

2-	**Which of the following is correct? (FINAL 31, FINAL 32)**
A-	Increased body temperature leads to increased cutaneous blood flow.
B-	A decrease in body temperature leads to increased cutaneous blood flow.
C-	During exercise renal blood flow increases.
D-	During exercise total peripheral resistance increases.

3-	**Regarding sweating? (FINAL 31)**
A-	Heat loss through sweating is controlled by parasympathetic system
B-	The efficacy of sweating is increased in humid atmosphere
C-	Profuse sweating can lead to heat stroke
D-	Sweating isa passive sweating

4-	**When one is exposed to cold all the following is true EXCEPT? (FINAL 31, 2ND TRIAL 32)**
A-	Decrease of the appetite
B-	Excessive secretion of Catecholamines
C-	Increase of muscle tone and shivering
D-	Ingestion of protein food is preferred due to its high SDA

5-	**Initially in the fever which of the following state is TRUE? (FINAL 31)**
A-	The body loses its thermoregulatory capability
B-	Body temperature continues to decrease
C-	Neurocontrol of peripheral vasculature is lost
D-	The hypothalamic set point temperature has been reset at a higher level

6-	**On exposure to hot? (FINAL 31, 2ND TRIAL 32)**		
A-	Sweating is inhibited	B-	Cutaneous vasodilatation occurs
C-	Increased appetite	D-	None of the above is correct

7-	**On exposure to cold? (FINAL 32)**
A-	Catecholamine secretion is inhibited
B-	The muscle tone increases and shivering may occur
C-	The parasympathetic is activated causing vasoconstriction & piloerection
D-	Cutaneous vasodilation occurs

8-	**in cold exposure sympathetic system simulates (MID 34)**		
A-	anterior hypothalamus	B-	posterior hypothalamus

9-	**in cold exposure sympathetic system simulates (MID 34)**		
A-	Shivering is inhibited	B-	Sweating is increased
C-	Lipolysis is stimulated	D-	Vasodilation of skin blood vessels

10-	**The human body can heat looses by... (MID 34)**
A-	Sweat evaporation

11-	**In fever: (FINAL 34)**
A-	Pyrogens rise the set point and make shivering

12-	**On exposure to hot weather? (2ND TRIAL 35)**
A-	Sweating is increased

13-	**Most of heat loss from the body occurs through? (2ND TRIAL 35)**		
A-	Skin	B-	Respiratory system
C-	Urine excretion	D-	Defecatio

14-	**When Body Temperature Increases.... (FINAL 37)**
A-	Inhibiting sympathetic centers in anterior hypothalamus
B-	Stimulate parasympathetic centers in anterior hypothalamus
C-	Inhibiting sympathetic centers in posterior hypothalamus

15-	**The metabolic rate is increased by all of the following except: (Formative 36)**		
A-	Rise in body temperature	B-	Increased sympathetic outflow.
C-	Rise in environmental temperature	D-	Hyperfunction of the thyroid gland.

16-	**Responses to cold exposure include: (Formative 36)**		
A-	Cutaneous vasoconstriction.	B-	Decreased cardiac output
C-	Increased cardiac output	D-	Opening of cutaneous vascular shunts.

17-	Which of the following structures contains the center that stimulates sweat secretion? (2ND TRIAL 37)
A-	Anterior hypothalamus
B-	Posterior hypothalamus
C-	Thalamus
D-	Medulla oblongata

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
D	A	C	A	D	B	B	B	C	A
11	12	13	14	15	16	17	18	19	20
A	A	A	C	C	A	A	-	-	-

Department Qs

1-	On exposure to cold: (SNU Bank, Dr. Mohamed Ashraf)
A-	The muscle tone increases and shivering may occur.
B-	There is excessive secretion of catecholamines.
C-	The sympathetic N'S. is activated causing, cutaneous V'C. and piloerection.
D-	Ingestion of protein food is preferred due to its high SDA.
E-	All of the above.

2-	The only heat loss mechanism that can occur in all environmental temperatures (SNU Bank, Dr. Mohamed Ashraf)						
A-	Radiation.	B-	Evaporation.	C-	Conduction.	D-	Convection

3-	At an environmental temperature 38°Cand humidity 20%, heat loss occurs by (SNU Bank, Dr. Mohamed Ashraf)						
A-	Conduction.	B-	Radiation.	C-	Convection.	D-	Evaporation.

4-	When the body temp. is lower than the atmospheric temp, heat loss occurs by: (SNU Bank, Dr. Mohamed Ashraf)		
A-	Radiation.	B-	Conduction.
C-	Convection.	D-	None of the above

5-	When the atmospheric temp is 15"Cheat loss occurs by (SNU Bank, Dr. Mohamed Ashraf)		
A-	Radiation and conduction.	B-	Increased urination and defecation.
C-	Evaporation of sweat.	D-	Increased respiration.

6-	**On exposure toa hot humid atmosphere: (SNU Bank, Dr. Mohamed Ashraf)**
A-	Sweat secretion is inhibited.
B-	V'D. Occurs in the skin blood vessels.
C-	Sweat evaporation increases
D-	Na' is retained in the body.
E-	The muscle tone and secretion of thyroxine tend to increase.

7-	**Heat stroke (hyperthermia) is primarily caused by: (SNU Bank, Dr. Mohamed Ashraf)**
A-	Excessive heat gain by conduction.
B-	Excessive heat production in the body.
C-	Inability to evaporate sweat due to excessive humidity.
D-	Inability to lose heat by radiation and other non-evaporative mechanisms.
E-	Too much clothing

8-	**Elevation of the body temperature to dangerous levels (hyperpyrexia): (SNU Bank, Dr. Mohamed Ashraf)**
A-	Is often produced as a result of excessive heat production.
B-	Is mainly due to irregularity of the central thermoregulatory mechanisms and development of vicious circles.
C-	Is fatal if the body temperature rises 3°Cabove the normal set point.
D-	Leads to death mainly due to renal failure and liver damage.

9-	**About the sweat glands: (SNU Bank, Dr. Mohamed Ashraf)**
A-	They are 5 types.
B-	The apocrine glands are responsible for temperature regulation.
C-	The eccrine glands secrete a hypertonic solution.
D-	The apocrine glands receive sympathetic cholinergic nerves.
E-	The secretion of the eccrine glands can be blocked by atropine.
F-	The apocrine glands are not responsive to the oxytocin hormone.

10-	**About the endogenous pyrogen, the following is true except (SNU Bank, Dr. Mohamed Ashraf)**
A-	It isa protein produced by monocytes and macrophages that causes fever.
B-	It is released by the effect of endotoxins released from pathogenic bacteria.
C-	It resets the hypothalamic thermostat toa higher set point than normal.
D-	It acts by increasing the local release of histamine in the hypothalamus.
E-	It increases the heat production mechanisms specially by inducing shivering
F-	It decreases the heat loss from the body by inducing cutaneous V'C.

11-	Which of the following statements is wrong (SNU Bank, Dr. Mohamed Ashraf)
A-	The endogenous pyrogen causes fever accompanied by a subjective cold sensation
B-	Aspirin produces an antipyretic effect by inhibiting prostaglandin synthesis.
C-	Fever is sometimes beneficial.
D-	The endogenous pyrogen activates the heat antidrop mechanisms.
E-	Hyperpyrexia is dangerous because it leads to heart failure.

12-	Concerning sweat, all the following is true except: (SNU Bank, Dr. Mohamed Ashraf)
A-	The sweat secretion center is located in the anterior hypothalamic nuclei.
B-	Its secretion starts at an atmospheric temperature of about 32-34°C
C-	it is an effective evaporative heat loss mechanism.
D-	Its evaporation occurs only in humid climates
E-	its excessive secretion leads to dehydration, salt loss and may be muscle cramps

13-	All the following statements are true except (SNU Bank, Dr. Mohamed Ashraf)
A-	In the comfortable zone of environmental temp., the body temp, is kept constant only by the non-evaporative heat loss mechanisms.
B-	Miner-scramps can be treated by giving salt tablets.
C-	The heat-loss center is located in the anterior hypothalamic nuclei and it produces temp. antirise responses mainly via physical processes.
D-	The heat-gain center is located in the posterior hypothalamic nuclei and it produces temp. antidrop responses mainly via chemical processes.
E-	The heat-loss center is stimulated in cold weather while the heat-gain center is stimulated in hot weather.

14-	About the body temperature: (SNU Bank, Dr. Mohamed Ashraf)
A-	Its regulation is given the priority in homeostasis.
B-	In very hot environments, it can be efficiently unaltered by wetting the skin by cold water.
C-	Evaporation of one ml of sweat leads to heat loss of about 0.6 Kcal.
D-	Atropine increases it while curare decreases it.
E-	All of the above are true.

15-	Which of the following is lower in sweat than in plasma (SNU Bank, Dr. Mohamed Ashraf)						
A-	NaCl	B-	K ions	C-	bicarbonate	D-	acids

16-	heat stroke can be treated by (SNU Bank, Dr. Mohamed Ashraf)						
A-	aspirin	B-	drinking alcohol	C-	ice bath	D-	pyrogens

17-	Vasodilatation of skin blood vessels that happens during exposure to hot weather is done by (SNU Bank, Dr. Mohamed Ashraf)		
A-	stimulation of sympathetic nerves	B-	stimulation of parasympathetic nerves
C-	inhibition of sympathetic nerves	D-	inhibition of parasympathetic nerves

18-	which of the following occurs during exposure to hot weather (SNU Bank, Dr. Mohamed Ashraf)		
A-	increase heat loss	B-	increase heat production
C-	stimulation of antidrop mechanisms	D-	inhibition of antirise mechanisms

19-	the centers for regulation of body temperature are present in (SNU Bank, Dr. Mohamed Ashraf)						
A-	hippocampus	B-	hypothalamus	C-	skin	D-	skeletal muscles

20-	most of heat loss from the body occurs through (SNU Bank, Dr. Mohamed Ashraf)		
A-	Skin	B-	urine excretion
C-	respiratory system	D-	defecation

21-	On exposure to cold? (SNU Bank)
A-	Muscle tone increases and shivering may occur
B-	Excessive secretion of catecholamines
C-	Sympathetic activated -> vasoconstriction & piloerection
D-	All the above

22-	Which is correct? (SNU Bank)
A-	Increased Body temp -> Increased Cutaneous blood flow
B-	Decreased Body temp -> Increased Cutaneous blood flow
C-	During exercise, renal blood flow increases
D-	During exercise, TPR increases

23-	Regarding sweating? (SNU Bank)		
A-	Controlled by parasympathetic	B-	Efficacy increased in humid atmosphere
C-	Profuse sweating can lead to heat stroke	D-	Sweating is passive

24-	On cold exposure, all true EXCEPT? (SNU Bank)		
A-	Decrease of appetite	B-	Excessive secretion of catecholamines
C-	Increase of muscle tone and shivering	D-	Protein food preferred due to high SDA

25-	Initially in fever, which is TRUE? (SNU Bank)		
A-	Body loses thermoregulatory capability	B-	Body temperature continues to decrease
C-	Neurocontrol of vasculature is lost	D-	Hypothalamic set point is reset at higher level

26-	On exposure to hot? (SNU Bank)		
A-	Sweating is inhibited	B-	Cutaneous vasodilatation occurs
C-	Increased appetite	D-	None of the above

27-	Which of the following is correct? (SNU Bank)
A-	Increased body temperature leads to increased cutaneous blood flow
B-	Decreased body temperature leads to increased cutaneous blood flow
C-	During exercise renal blood flow increases
D-	During exercise total peripheral resistance increases

28-	On exposure to cold? (SNU Bank)
A-	Catecholamine secretion is inhibited
B-	Muscle tone increases and shivering may occur
C-	Parasympathetic causes vasoconstriction & piloerection
D-	Cutaneous vasodilation occurs

29-	When one is exposed to cold all the followings are true EXCEPT? (SNU Bank)
A-	Decrease of the appetite
B-	Excessive secretion of Catecholamines
C-	Increase of muscle tone and shivering
D-	Ingestion of protein food is preferred due to its high SDA

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
E	B	D	D	A	B	C	B	E	D
11	12	13	14	15	16	17	18	19	20
E	D	E	D	A	C	C	A	B	A
21	22	23	24	25	26	27	28	29	30
D	A	C	A	D	B	A	B	A	B
31	32	33	34	35	36	37	38	39	40
A	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	**Cold-blooded animals are characterized by: (M3WAN - Core)**
A-	Maintaining a constant body temperature despite changes in atmospheric temperature
B-	Having a hypothalamic set point identical to warm-blooded animals
C-	Generating heat exclusively through shivering
D-	Having no mechanism for regulation of body temperature, which rises or falls according to atmospheric temperature (e'g', frogs and fishes)

2-	**Warm-blooded animals, such as mammals and man, are characterized by: (M3WAN - Core)**
A-	Body temperature that rises or falls passively with atmospheric temperature
B-	Complete absence of any heat production mechanism
C-	The ability to keep body temperature constant despite changes in atmospheric temperature
D-	Dependence entirely on external heat sources

3-	**Body temperature is conventionally measured using a: (M3WAN - Core)**		
A-	Sphygmomanometer	B-	Medical mercury thermometer
C-	Cathodal ray oscilloscope	D-	Spectrophotometer

4-	**The normal range of body temperature is approximately: (M3WAN - Core)**							
A-	36.5 to 37.2°C	B-	34.0 to 35.5°C	C-	38.0 to 39.5°C	D-	35.0 to 36.0°C

5-	**Axillary temperature measurement compared with oral temperature is: (M3WAN - Core)**		
A-	0.5°Chigher than the oral temperature	B-	Identical to the oral temperature
C-	0.5°Clower than the oral temperature	D-	1°Clower than the oral temperature

6-	**Rectal temperature measurement compared with oral temperature is: (M3WAN - Core)**		
A-	0.5°Clower than the oral temperature	B-	Identical to the oral temperature
C-	1°Chigher than the oral temperature	D-	0.5°Chigher than the oral temperature

7-	**Axillary or rectal measurement of body temperature is indicated in all of the following EXCEPT: (M3WAN - Core)**		
A-	Infants	B-	A fully cooperative healthy adult at rest
C-	Patients in coma	D-	Patients under anesthesia

8-	**Physiological variations in body temperature occur with all of the following EXCEPT: (M3WAN - Core)**		
A-	Complete cessation of metabolic activity	B-	Diurnal variation
C-	Exercise	D-	Ovulation in women

9-	An abnormal increase in body temperature is called: (M3WAN - Core)		
A-	Hypothermia	B-	Euthermia
C-	Normothermia	D-	Hyperthermia or fever

10-	An abnormal decrease in body temperature is called: (M3WAN - Core)						
A-	Hyperthermia	B-	Pyrexia	C-	Hypothermia	D-	Fever

11-	Regulation of body temperature depends on the balance between: (M3WAN - Core)
A-	Blood glucose and insulin levels
B-	Heat produced in the body and heat lost from the body
C-	Water intake and water excretion
D-	Sodium intake and sodium excretion

12-	The major portion of body heat production, called the heat of metabolism, results from: (M3WAN - Core)		
A-	Metabolism of foodstuffs	B-	Radiation from the environment
C-	Evaporation of sweat	D-	Convection currents in air

13-	Heat is produced in the muscles: (M3WAN - Core)		
A-	Only during vigorous activity, never at rest	B-	Only at rest, never during activity
C-	Only during sleep	D-	Both at rest and during activity

14-	Hormones that increase heat production include: (M3WAN - Core)		
A-	Insulin and glucagon	B-	Estrogen and progesterone
C-	Thyroxine and adrenaline	D-	Calcitonin and parathormone

15-	Radiation of heat from the environment to the body, as a source of heat gain, occurs when: (M3WAN - Hard)
A-	The environmental temperature is higher than the body temperature
B-	The environmental temperature is lower than the body temperature
C-	The environmental and body temperatures are equal
D-	The body is submerged in water

16-	Shivering as a mechanism of heat production is best described as: (M3WAN - Core)
A-	A pathological mechanism seen only in fever
B-	A compensatory physiological mechanism involving contraction/twitching of muscles, producing enormous heat, as during exposure to cold
C-	A mechanism that decreases heat production

D-	A voluntary muscular activity unrelated to temperature regulation

17-	One of the tissues involved in heat production, particularly important in infants, is: (M3WAN - Core)		
A-	White adipose tissue exclusively	B-	Bone marrow
C-	Cartilage	D-	Brown fat tissue

18-	Maximum heat loss from the body occurs through the: (M3WAN - Core)		
A-	Respiratory system	B-	Kidneys
C-	Skin	D-	Gastrointestinal tract

19-	Heat lost from the body surface to other objects such asa chair or bed, by direct contact, is called: (M3WAN - Core)
A-	Radiation, accounting for about 60% of heat loss
B-	Conduction, accounting for about 3% of heat loss
C-	Convection, accounting for about 15% of heat loss
D-	Evaporation, accounting for about 22% of heat loss

20-	The largest single route of heat loss from the body, accounting for approximately 60%, is: (Important M3WAN - Core)
A-	Radiation, by infrared electromagnetic radiation to surrounding objects
B-	Conduction
C-	Convection
D-	Evaporation (insensible perspiration)

21-	Heat carried away from the body surface by air currents is called: (M3WAN - Core)
A-	Conduction, accounting for about 3% of heat loss
B-	Radiation, accounting for about 60% of heat loss
C-	Panting
D-	Convection, accounting for about 15% of heat loss

22-	Continuous evaporation of water from the skin and lungs, of which we are normally unaware, is called: (M3WAN - Core)
A-	Sensible sweating, accounting for about 60% of heat loss
B-	Conduction
C-	Insensible perspiration, accounting for about 22% of heat loss
D-	Panting

23-	In animals such as dogs, which lack sweat glands, heat is lost mainly through evaporation of water from the lungs and saliva by means of: (M3WAN - Core)		
A-	Conduction	B-	Panting (rapid shallow breathing)
C-	Radiation	D-	Shivering

24-	The hypothalamic set point for body temperature under normal physiological conditions is: (M3WAN - Core)						
A-	37°C	B-	35°C	C-	39°C	D-	40°C

25-	The hypothalamus regulates body temperature through two centers, namely the: (M3WAN - Core)
A-	Feeding center and satiety center
B-	Preoptic and arcuate nuclei only, with no gain/loss distinction
C-	Anterior and posterior pituitary centers
D-	Heat loss center and heat gain center

26-	The heat loss center is situated in the: (M3WAN - Core)		
A-	Posterior hypothalamic nucleus	B-	Arcuate nucleus
C-	Preoptic nucleus of the anterior hypothalamus	D-	Paraventricular nucleus

27-	Neurons in the preoptic nucleus that are heat-sensitive are called: (M3WAN - Core)						
A-	Chemoreceptors	B-	Thermoreceptors	C-	Baroreceptors	D-	Osmoreceptors

28-	Stimulation of the preoptic nucleus results in: (Important M3WAN - Core)		
A-	Cutaneous vasodilatation and sweating	B-	Cutaneous vasoconstriction and shivering
C-	No change in skin blood flow	D-	Increased metabolic heat production

29-	Removal or lesion of the preoptic nucleus (heat loss center) results in: (Important M3WAN - Core)
A-	A decrease in body temperature (hypothermia)
B-	No change in body temperature
C-	Immediate death from hypothermia
D-	An increase in body temperature (hyperthermia)

30-	The heat gain (heat production) center is situated in the: (M3WAN - Core)		
A-	Preoptic nucleus of the anterior hypothalamus	B-	Ventromedial nucleus
C-	Posterior hypothalamic nucleus	D-	Suprachiasmatic nucleus

31-	**Stimulation of the posterior hypothalamic nucleus (heat gain center) causes: (Important M3WAN - Core)**		
A-	Sweating	B-	Shivering
C-	Cutaneous vasodilatation	D-	Inhibition of the sympathetic nervous system

32-	**Removal or lesion of the posterior hypothalamic nucleus (heat gain center) leads to: (Important M3WAN - Core)**		
A-	A fall in body temperature	B-	Arise in body temperature
C-	No change in body temperature	D-	Immediate hyperthermia

33-	**When body temperature increases, the heat loss center brings it back to normal by promoting heat loss, which includes: (Important M3WAN - Core)**
A-	Stimulating sympathetic centers to cause cutaneous vasoconstriction
B-	Stimulating shivering
C-	Increasing metabolic (chemical) reactions
D-	Increasing sweat secretion and inhibiting sympathetic centers in the posterior hypothalamus to cause cutaneous vasodilatation

34-	**When body temperature increases, the heat loss center also brings it back to normal by prevention of heat production, achieved by: (M3WAN - Core)**
A-	Stimulating shivering
B-	Stimulating adrenaline secretion
C-	Inhibiting mechanisms involved in heat production, such as shivering and chemical (metabolic) reactions
D-	Increasing muscular activity

35-	**When body temperature decreases, prevention of heat loss is achieved mainly by: (Important M3WAN - Core)**
A-	Sympathetic centers in the posterior hypothalamus causing cutaneous vasoconstriction, decreasing blood flow to the skin
B-	Sympathetic centers causing cutaneous vasodilatation
C-	Increased sweat secretion
D-	Inhibition of the heat gain center

36-	**When body temperature decreases, promotion of heat production is achieved by two ways, namely: (Important M3WAN - Core)**
A-	Sweating and cutaneous vasodilatation
B-	Shivering (via the primary motor center for shivering in the posterior hypothalamus) and increased metabolic reactions (via sympathetic stimulation of adrenaline and noradrenaline secretion)
C-	Inhibition of the heat gain center and cutaneous vasodilatation

D-	Panting and conduction

37-	**Heat loss through sweating is primarily controlled by the: (Important M3WAN - Core)**		
A-	Parasympathetic nervous system	B-	Somatic nervous system
C-	Enteric nervous system	D-	Sympathetic nervous system

38-	**The efficacy of sweating as a heat-loss mechanism is: (M3WAN - Core)**		
A-	Increased in a humid atmosphere	B-	Unaffected by atmospheric humidity
C-	Decreased in a humid atmosphere	D-	Maximal only in cold, dry conditions

39-	**When exposed to cold, all of the following occur EXCEPT: (Important M3WAN - Hard)**
A-	Excessive secretion of catecholamines
B-	Decreased appetite
C-	Increased muscle tone and shivering
D-	Preference for protein-rich food due to its high specific dynamic action

40-	**In the initial stage of fever, which of the following is TRUE? (Important M3WAN - Core)**
A-	The hypothalamic set point temperature has been reset at a higher level
B-	The body loses its thermoregulatory capability entirely
C-	Body temperature continues to decrease
D-	Neurocontrol of peripheral vasculature is completely lost

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	B	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	A	D	C	B	A	D	C
31	32	33	34	35	36	37	38	39	40
B	A	D	C	A	B	D	C	B	A

Metabolism

Pre Exams & Formative

1-	Abasal metabolic rate (BMR) of (-50%) means? (FINAL 31)		
A-	BMR = 50 Cal/m/hr	B-	BMR = 20 Cal/m/hr
C-	BMR is less than zero by 50 Calories	D-	BMR is less than zero by 20 Calories

2-	Deficiency of which of these hormones increases the BMR? (FINAL 31)								
A-	Thyroxine	B-	Adrenaline	C-	Cortisol	D-	TSH	E-	ADH

3-	Metabolism of 10 gm of carbohydrate inside the body gives ..... Calories? (FINAL 31)								
A-	4.1	B-	41	C-	410	D-	9.3	E-	5.3

4-	Complete oxidation of 1 gm of protein outside the body Releases energy? (FINAL 32)		
A-	More than 1 gm of fat	B-	Less than 1 gm of carbohydrate
C-	Equal to 1 gm of carbohydrate	D-	More than 1 gm of protein inside the body

5-	RQ of diabetic patient is always 0.7 because of? (FINAL 32)		
A-	High blood glucose level	B-	Dependence on fat metabolism
C-	Conversion of glucose into fat	D-	Obesity and lack of exercise

6-	BMR increases by which of the following? (FINAL 32)		
A-	People consuming high protein diet	B-	Dark races
C-	During sleep	D-	More fat tissue in the body

7-	Metabolism of 10 gm of fat produces? (2ND TRIAL 32)						
A-	9.3	B-	3.9	C-	39	D-	93

8-	Specific dynamic action mostly increases after ingestion of? (2ND TRIAL 32)						
A-	Proteins	B-	Carbohydrate	C-	Lipids	D-	Water

9-	The RQ of a patient with diabetes mellitus (FINAL 34)
A-	0.7

10-	Which of the following is true about carbohydrates? (FINAL 34)
A-	physiological heat value is more than physical heat value
B-	physical heat value and physiological heat value are 5.3
C-	physical heat value and physiological heat value are 4.1

11-	Respiratory quotient R'Q decreases in which of the following? (FINAL 35, 2ND TRIAL 35)						
A-	Diabetes mellitus	B-	muscular exercise	C-	Acidosis	D-	Febrile conditions

12-	All of the following decrease the RQ Except? (2ND TRIAL 35)						
A-	Diabetes mellitus	B-	Alkalosis	C-	Hypoventilation	D-	Muscular exercise

13-	The RQ is 0.7 when fat is oxidized due to: (2ND TRIAL 37)
A-	Having a low O_2 content in its molecule
B-	Increased CO_2 production and decreased O_2 consumption
C-	High O_2 content in its molecule
D-	Increased conversion of carbohydrates into fats

14-	Which of the following is correct regarding proteins? (2ND TRIAL 37)
A-	The physical heat value of protein is more than its physiological heat value
B-	The physical heat value of protein is equal to the physiological heat value of carbohydrate
C-	The physiological heat value of protein is 9.3 Calories/gram
D-	The physical heat value of protein is 4.1 Calories/gram

15-	All of the following lead to an increase in BMR EXCEPT: (2ND TRIAL 37)
A-	Hypothermia
B-	Hyperthyroidism
C-	Cold weather
D-	Pregnancy

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
B	E	B	D	B	A	D	A	A	C
11	12	13	14	15	16	17	18	19	20
A	D	A	A	A	-	-	-	-	-

Department Qs

1-	About the energy balance: (SNU Bank, Dr. Mohamed Ashraf)
A-	In normal adults, the energy intake is greater than the energy output.
B-	During growth, the energy intake is equal to the energy output.
C-	Witha negative energy balance, there is loss of the body weight.
D-	In obesity, there is increased protein content of the body.

2-	**Most of the energy released from catabolic processes: (SNU Bank, Dr. Mohamed Ashraf)**
A-	Appears as thermal (heat) energy.
B-	Is stored as high energy phosphate bonds.
C-	Is used by the kidneys to perform osmotic work.
D-	is used for building new protoplasm.

3-	**The physical heat value of: (SNU Bank, Dr. Mohamed Ashraf)**
A-	Food is measured by the respirometer.
B-	Carbohydrate is equal to its physiological heat value.
C-	Protein is 9.3 Kcal/gm.
D-	Fat is 5.3 Kcal/gm

4-	**The metabolism of 10 gm protein inside the body would produce: (SNU Bank, Dr. Mohamed Ashraf)**							
A-	53 Kcal.	B-	93 Kcal.	C-	410 Kcal.	D-	41 Kcal.

5-	**A respiratory quotient above one occurs in all the following cases except: (SNU Bank, Dr. Mohamed Ashraf)**
A-	Hyperventilation.
B-	Muscular exercise.
C-	Fevers and acidosis.
D-	Transformation of carbohydrate into fat.
E-	Gluconeogenesis using fat as a substrate to be converted into glucose.

6-	**The respiratory quotient (RO): (SNU Bank, Dr. Mohamed Ashraf)**
A-	Is the ratio of the volume of O2 consumed to that of CO2 produced.
B-	Rises in the second week of starvation.
C-	Is 0.7 when glucose is the metabolized substrate.
D-	Is about 1 in the brain.
E-	Is 1 in uncontrolled diabetic patients.

7-	**The excess R.Q. : (SNU Bank, Dr. Mohamed Ashraf)**
A-	is the R'Q. determined during sleep.
B-	Indicates the fuel used during a certain exercise.
C-	Determination requires estimation of the O2, consumption and CO, production during the recovery period only.
D-	Is about 1 ina prolonged exercise and 0.7 in short exercises.

8-	**About the BMR, all the following is true except: (SNU Bank, Dr. Mohamed Ashraf)**
A-	It is expressed in Kcal /square meter/hour.
B-	Normally, it is about 40 Kcal in young adult males.
C-	It represents the energy expenditure needed for activity of the vital organs
D-	It is slightly higher in males and is maximal in children at the age of 5 years
E-	It is measured immediately after dinner at an atmospheric temp about 10C.

9-	**The energy equivalent of oxygen is (SNU Bank, Dr. Mohamed Ashraf)**
A-	The amount of heat (Kcal) produced when one litre of O2 is consumed.
B-	Used for estimation of the metabolic rate by direct calorimetry.
C-	About 5 Kcal /Lin case of fats and 4.7 Kcal /Lin case of carbohydrates.
D-	About 4.8 Kcal /Lin case of proteins and 4.6 Kcal /Lin the post-absorptive
E-	Not equal to the heat value of O2.

10-	**About the specific dynamic action (SDA), all the following is true except: (SNU Bank, Dr. Mohamed Ashraf)**
A-	It is an increase in the metabolic rate after food intake.
B-	The SDA of proteins is about 30% and is due mainly to its metabolism in the liver
C-	It is generally due to the work done during digestion and absorption of food.
D-	Normally, it is about 6% for carbohydrates and 4% for fats.
E-	It is used for warming the body on exposure toa cold weather.

11-	**Ingestion of protein meals. (SNU Bank, Dr. Mohamed Ashraf)**
A-	Causes a marked increase in the metabolic rate.
B-	Is preferable in hot climate.
C-	Leads to decreased urea formation in the liver.
D-	Decreases the synthesis of essential amino acids in the liver.
E-	Causes the liver to derive its energy needs from oxidation of fatty acids.

12-	**After one week of starvation: (SNU Bank, Dr. Mohamed Ashraf)**
A-	The respiratory quotient becomes about 1.
B-	The blood glucose level is much decreased below the fasting level.
C-	The urine contains an excessive amount of NH, and its pH drops.
D-	The main source of energy is the liver glycogen.
E-	The serum lipid level is not significantly changed.

13-	**It is false that during starvation: (SNU Bank, Dr. Mohamed Ashraf)**
A-	The blood glucose level remains constant at the fasting level.
B-	There is lipaemia and ketosis.
C-	The plasma protein concentration decreases.
D-	The amount of the alkali reserve is decreased.
E-	The blood volume remains constant.

14-	**During starvation: (SNU Bank, Dr. Mohamed Ashraf)**
A-	The body gets energy first from carbohydrates then from fat and finally from proteins.
B-	Protein catabolism is unavoidable from the start.
C-	The blood levels of most electrolytes remain fairly constant.
D-	The brain & heart are little affected till the terminal stage.
E-	Signs of vitamin deficiency and acidosis develop as starvation advances.
F-	All of the above is true

15-	**Which of the following statements is true? (SNU Bank, Dr. Mohamed Ashraf)**
A-	The metabolic rate can be estimated if the R'Q. and O2 consumption are known.
B-	One gram of protein gives the same amount of energy whether it is oxidized in a bomb calorimeter or metabolized inside the body
C-	The specific dynamic action persists for about 2 days after food ingestion.
D-	The nitrogen balance is more negative in early than in late starvation.
E-	An increase in the body fat content causes reduction of the lean body mass.

16-	**When RQ value is 0.7, this indicates that the person uses ... as a source of energy (SNU Bank, Dr. Mohamed Ashraf)**		
A-	pure protein	B-	pure fat
C-	pure carbohydrate	D-	50% fat + 50% carbohydrate

17-	**Which of the following increases the RQ above one: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	diabetes mellitus	B-	hyperventilation
C-	alkalosis	D-	recovery from exercise

18-	**Basal metabolic rate is measured under basal conditions which include: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	Sleep	B-	after meals
C-	exercise	D-	postabsorptive state

19-	which of the following factors increases the basal metabolic rate (SNU Bank, Dr. Mohamed Ashraf)		
A-	old aged	B-	females
C-	atheletics	D-	people living in hot climate

20-	metabolic rate is increased much after ingestion of (SNU Bank, Dr. Mohamed Ashraf)						
A-	protein	B-	fat	C-	carbohydrate	D-	water

21-	specific dynamic action is due to (SNU Bank, Dr. Mohamed Ashraf)		
A-	digestion of food	B-	absorption of food
C-	the act of swallowing and mastication	D-	the chemical reaction in the liver

22-	specific dynamic action is absent when the external temperature is (SNU Bank, Dr. Mohamed Ashraf)						
A-	37	B-	33	C-	3	D-	13

23-	BMR of (-50%) means? (SNU Bank)		
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32-	All of the following decrease the RQ EXCEPT? (SNU Bank)						
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11	12	13	14	15	16	17	18	19	20
A	C	E	D	A	B	B	D	C	A
21	22	23	24	25	26	27	28	29	30
D	C	B	E	B	D	B	A	D	A
31	32	33	34	35	36	37	38	39	40
A	D	-	-	-	-	-	-	-	-

M3WAN Bank

1-	The balance between anabolism (energy intake) and catabolism (energy output) normally results in: (M3WAN - Core)		
A-	Progressive weight gain regardless of diet	B-	Constant body weight
C-	Progressive weight loss regardless of diet	D-	Random, unpredictable weight fluctuations

2-	A state of predominant anabolism, as occurs in children, results in: (M3WAN - Core)		
A-	Decreased body weight	B-	No change in body weight
C-	Increased body weight	D-	Immediate catabolic crisis

3-	A state of predominant catabolism, as occurs in hyperthyroidism, results in: (M3WAN - Core)		
A-	Increased body weight	B-	Decreased body weight
C-	No change in body weight	D-	Fluid retention only

4-	Anabolism is defined as: (M3WAN - Core)
A-	The conversion of large molecules into small molecules with production of energy
B-	The conversion of small molecules into large molecules with consumption of energy ,e'g', amino acids into proteins

C-	The complete breakdown of nutrients into CO2 and water
D-	A process that never occurs in growing children

5-	**Catabolism is defined as: (M3WAN - Core)**
A-	The conversion of small molecules into large molecules with consumption of energy
B-	A process exclusive to the liver
C-	The conversion of large molecules into small molecules with production of energy,e'g', glucose into CO2, H2O, and energy
D-	A process that only occurs during starvation

6-	**Metabolism is best defined as: (M3WAN - Core)**
A-	All chemical reactions occurring in living cells that make it possible to continue living
B-	Only the reactions involved in digestion
C-	Only the reactions involved in protein synthesis
D-	The process of hormone secretion exclusively

7-	**Of the total energy liberated by catabolism, the fraction stored as ATP, creatine phosphate, and glycogen approximates: (Important M3WAN - Core)**						
A-	7%	B-	50%	C-	93%	D-	27%

8-	**Besides being stored in energy stores, energy liberated by catabolism is also: (M3WAN - Core)**		
A-	Liberated as heat	B-	Entirely lost without any physiological use
C-	Converted directly into structural fat only	D-	Used exclusively for DNA replication

9-	**Energy is used directly by cells for certain kinds of work, including all of the following EXCEPT: (M3WAN - Core)**
A-	Membrane transport (active transport)
B-	Muscle contraction
C-	Complete cessation of all cellular activity
D-	Glandular secretion, such as HCl secretion by gastric glands

10-	**A calorie is defined as: (M3WAN - Core)**
A-	The amount of heat necessary to raise the temperature of 1 kg of water by 1°C, from 15°Cto 16°C
B-	The amount of heat necessary to raise the temperature of 1g of water by 10°C
C-	The energy content of 1 gram of carbohydrate
D-	The energy required to synthesize 1 mole of ATP

11-	One kilocalorie is equal to: (M3WAN - Core)						
A-	1 calorie	B-	100 calories	C-	1000 calories	D-	10 calories

12-	Physical heat value of a foodstuff is measured by complete oxidation of the food outside the body, using a: (M3WAN - Core)
A-	Spectrophotometer
B-	Bomb calorimeter, in which the food is ignited by an electric spark
C-	Cathodal ray oscilloscope
D-	Direct calorimeter measuring only body heat loss

13-	Physiological heat value ofa foodstuff refers to the amount of heat produced by: (M3WAN - Core)
A-	Complete oxidation of the food outside the body
B-	Partial oxidation of the food outside the body
C-	Complete oxidation of the food inside the body
D-	Storage of the food as glycogen

14-	The physical and physiological heat values of carbohydrate are both approximately: (M3WAN - Core)						
A-	9.3 Cal/g	B-	4.1 Cal/g	C-	5.3 Cal/g	D-	7.0 Cal/g

15-	The physical and physiological heat values of fat are both approximately: (M3WAN - Core)						
A-	4.1 Cal/g	B-	5.3 Cal/g	C-	9.3 Cal/g	D-	6.5 Cal/g

16-	Regarding the heat value of protein: (Important M3WAN - Core)
A-	The physical value (5.3 Cal/g) is greater than the physiological value (4.1 Cal/g only)
B-	The physical and physiological values are identical, as with carbohydrate and fat
C-	The physiological value exceeds the physical value
D-	Protein has no measurable heat value

17-	The physiological heat value of protein is lower than its physical heat value mainly because: (Important M3WAN - Core)
A-	Protein is never fully absorbed from the gut
B-	The nitrogen component of protein is not oxidized in the body (its oxidation would produce dangerous amounts of nitric acid) and is instead converted to urea and excreted in urine
C-	Protein is stored entirely as fat before oxidation
D-	Protein contains no carbon atoms

18-	The respiratory quotient (RQ) is defined as: (Important M3WAN - Core)
A-	The ratio between the volume of O2 consumed and the volume of CO2 produced
B-	The ratio between the volume of CO2 produced and the volume of O2 consumed in the same period of metabolism
C-	The total volume of air breathed per minute
D-	The ratio of heart rate to respiratory rate

19-	The respiratory quotient (RQ) is useful as all of the following EXCEPT: (M3WAN - Core)
A-	An index of the nature of the substances being oxidized
B-	A tool for calculating the metabolic rate
C-	Evidence for the transformation of one food substance into another
D-	A direct measurement of blood glucose concentration

20-	The RQ of carbohydrate is: (Important M3WAN - Core)						
A-	0.7	B-	0.8	C-	1.0	D-	0.85

21-	The RQ of protein is: (M3WAN - Core)						
A-	1.0	B-	0.7	C-	0.85	D-	0.8

22-	The RQ of fat is: (Important M3WAN - Core)						
A-	1.0	B-	0.8	C-	0.7	D-	0.85

23-	The RQ of a mixed diet approximates: (M3WAN - Core)						
A-	1.0	B-	0.7	C-	0.85	D-	0.5

24-	The RQ of carbohydrate equals 1.0 because: (M3WAN - Hard)
A-	Carbohydrate contains no hydrogen atoms
B-	Carbohydrate molecules contain enough oxygen to oxidize all their hydrogen atoms, so extra atmospheric O2 is used only to oxidize carbon (forming CO2 in equal moles to O2 consumed)
C-	Carbohydrate is never fully oxidized in the body
D-	Carbohydrate oxidation produces no CO2 at all

25-	The RQ of fat is lower than that of carbohydrate mainly because: (M3WAN - Hard)
A-	Fat molecules contain less oxygen relative to their hydrogen and carbon content, requiring more atmospheric O2 for complete oxidation
B-	Fat contains more oxygen than carbohydrate
C-	Fat is never oxidized inside the body

D-	Fat oxidation produces more CO2 than O2 consumed

26-	The RQ of the brain approximates 1.0 (0.97-0.99), indicating that: (Important M3WAN - Core)		
A-	Fat is the brain-sprincipal fuel	B-	The brain does not consume oxygen
C-	Glucose is the brain-sprincipal fuel	D-	Protein is the brain-sprincipal fuel

27-	The RQ in diabetes mellitus approximates 0.7, indicating that: (Important M3WAN - Core)		
A-	Carbohydrate is the chief source of energy	B-	Fat is the chief source of energy
C-	Protein is the chief source of energy	D-	No substrate is being oxidized

28-	The metabolic rate can be calculated as: (M3WAN - Core)
A-	RQ multiplied by body weight
B-	The O2 heat value (obtained from RQ using metabolic tables) multiplied by the O2 consumed
C-	CO2 produced divided by body surface area
D-	Body temperature multiplied by respiratory rate

29-	The O2 heat value is defined as: (M3WAN - Core)
A-	The amount of heat in calories resulting from oxidation ofa food substance using one liter of O2
B-	The amount of O2 required to produce one calorie of heat
C-	The RQ multiplied by two
D-	The heat produced by burning one gram of fat

30-	Transformation of carbohydrates into fats (lipogenesis) is evidenced by an RQ that is: (Important M3WAN - Core)						
A-	Below unity	B-	Exactly zero	C-	Unaffected	D-	Above unity

31-	Transformation of fats or proteins into carbohydrates (gluconeogenesis) is evidenced by an RQ that is: (Important M3WAN - Core)		
A-	Above unity	B-	Below unity
C-	Exactly 1.0 at all times	D-	Unrelated to substrate transformation

32-	The RQ approaches unity immediately after a meal mainly because: (M3WAN - Core)
A-	The main food substance being metabolized at that time is carbohydrate
B-	Fat becomes the main substrate immediately after eating
C-	Protein becomes the main substrate immediately after eating
D-	Metabolism ceases temporarily after eating

33-	**In the early stages of severe muscular exercise, the RQ can rise as high as 2, mainly due to: (Important M3WAN - Core)**
A-	Complete cessation of CO2 production
B-	Accumulation of lactic acid, which when buffered by NaHCO3 produces CO2 without consuming O2, together with hyperventilation
C-	A sudden drop in body temperature
D-	Complete inhibition of the respiratory center

34-	**Metabolic acidosis tends to raise the RQ above unity mainly because: (Important M3WAN - Core)**
A-	The excess acids are buffered by NaHCO3, and hyperventilation further increases CO2 output relative to O2 uptake
B-	Acidosis abolishes CO2 production entirely
C-	Acidosis stops O2 consumption completely
D-	Acidosis has no relation to gas exchange

35-	**Hyperventilation tends to raise the RQ above unity mainly because: (M3WAN - Core)**
A-	CO2 is washed out of the body faster than O2 is taken up
B-	O2 uptake exceeds CO2 washout
C-	Both O2 uptake and CO2 output stop
D-	Hyperventilation has no effect on gas exchange ratios

36-	**About 8-10 hours after a meal, the RQ approaches 0.7 mainly because: (M3WAN - Core)**
A-	Carbohydrate stores are already used up, and the main substrate being metabolized becomes fat
B-	Carbohydrate remains the main substrate
C-	Protein becomes the exclusive substrate
D-	Metabolism stops entirely

37-	**Recovery from muscular exercise is associated with an RQ below unity mainly because: (Important M3WAN - Core)**
A-	Lactic acid is oxidatively removed and CO2 is used for bicarbonate reformation, so O2 consumption rises while CO2 production falls
B-	Lactic acid accumulates further
C-	O2 consumption falls while CO2 production rises
D-	Bicarbonate is broken down, releasing CO2

38-	**Metabolic alkalosis and hypoventilation tend to lower the RQ below unity mainly due to: (M3WAN - Core)**
A-	Retention of CO2 in the body
B-	Excessive CO2 washout
C-	Increased O2 consumption without any change in CO2
D-	Complete cessation of metabolism

39-	**Starvation and diabetes mellitus are both associated with an RQ below unity, in diabetes mainly because of: (Important M3WAN - Core)**
A-	Decreased carbohydrate metabolism, increased fat metabolism, and formation of oxygen-rich ketone bodies from oxygen-poor fat
B-	Increased carbohydrate metabolism exclusively
C-	Complete absence of any metabolic activity
D-	Increased protein synthesis exclusively

40-	**Excess RQ is defined as: (M3WAN - Core)**
A-	The RQ measured only at complete rest
B-	The RQ of the excess metabolism due to muscular exercise
C-	The RQ measured only during sleep
D-	The difference between carbohydrate and fat RQ values

41-	**The formula for excess RQ, (A'a)/(B'b), uses A..... and a..... to represent: (M3WAN - Hard)**
A-	Total O2 consumed during exercise/recovery, and O2 consumed during a similar resting period
B-	Body weight before and after exercise
C-	Total CO2 produced during exercise/recovery, and CO2 produced during a similar resting period
D-	Heart rate before and after exercise

42-	**An excess RQ of unity, without any subsequent decline, indicates that the exercise was: (Important M3WAN - Core)**
A-	Of short duration, with energy derived mainly from carbohydrate oxidation
B-	Of long duration, with energy derived from fat only
C-	Of short duration, with energy derived mainly from fat oxidation
D-	Of indeterminate duration and fuel source

43-	**An excess RQ that is below unity from the very start of exercise indicates that: (Important M3WAN - Core)**
A-	Carbohydrate is the only available fuel
B-	The only available fuel is fat, as occurs in diabetes and starvation
C-	Protein is the only available fuel
D-	No fuel is being oxidized at all

44-	**The basal metabolic rate (BMR) is defined as: (Important M3WAN - Core)**
A-	The amount of energy production in unit time (one hour) under basal conditions
B-	The maximal possible energy expenditure of the body
C-	The energy expended only during vigorous exercise
D-	The energy required for digestion alone

45-	**The BMR represents the energy required to perform all essential activities of body systems (CNS, heart, respiration, kidneys, and skeletal muscle tone),i'e', it represents: (M3WAN - Core)**
A-	A wasteful and unnecessary energy expenditure
B-	The unavoidable cost of life
C-	Only the energy needed for muscular exercise
D-	The energy needed exclusively for growth

46-	**Although it represents the unavoidable cost of life, the BMR is NOT the absolute minimal energy expenditure, because it: (M3WAN - Hard)**
A-	Can be further reduced by factors such as sleep
B-	Is always fixed and cannot change under any circumstance
C-	Increases automatically with age
D-	Is identical in all individuals regardless of condition

47-	**One of the standard basal conditions required for accurate BMR measurement is the postabsorptive state, meaning the measurement is taken: (Important M3WAN - Core)**
A-	Immediately after a meal
B-	12-14 hours after the last meal, to avoid the stimulating effect of food (specific dynamic action) on metabolic rate
C-	During active digestion
D-	Only after a 24-hour fast

48-	**Another standard basal condition for BMR measurement isa comfortable external temperature, which is approximately: (M3WAN - Hard)**
A-	0-5°Cfora dressed person
B-	35-40°Cfora naked person
C-	20-25°Cfora dressed person, and 28-31°Cwhen naked
D-	Identical regardless of clothing

49-	**BMR is measured using: (M3WAN - Core)**		
A-	A direct or indirect calorimeter	B-	A sphygmomanometer
C-	A cathodal ray oscilloscope	D-	An electrocardiograph

50-	**The normal BMR in adults approximates: (Important M3WAN - Core)**		
A-	100 Calories/m2/hr ± 15%	B-	10 Calories/m2/hr ± 15%
C-	70 Calories/m2/hr ± 15%	D-	40 Calories/m2/hr ± 15%

51-	**Clinically, BMR is expressed as: (M3WAN - Core)**
A-	An absolute value only, never compared to normal
B-	A percentage deviation from the normal standard
C-	A ratio to body weight only
D-	A fixed constant for all patients

52-	**Regarding the effect of age on BMR, the highest BMR value in life occurs at approximately: (Important M3WAN - Core)**		
A-	2 to 5 years of age	B-	Birth (newborn period)
C-	Puberty	D-	20 years of age

53-	**After the age of 20 years, BMR: (M3WAN - Core)**
A-	Increases progressively due to increased muscle mass
B-	Remains completely constant throughout life
C-	Decreases by about 1 calorie for every 10 years, due to decreased muscle mass being replaced by fat
D-	Doubles every decade

54-	**Regarding the effect of sex on BMR: (Important M3WAN - Core)**
A-	BMR in females is lower than in males by about 7%, mainly due to greater fat stores (which have low metabolic activity) rather than hormonal differences
B-	BMR in females is higher than in males due to hormonal differences
C-	There is no difference in BMR between males and females
D-	The difference is purely hormonal and absent in children

55-	**Regarding the effect of physical habits on BMR: (M3WAN - Core)**
A-	Athletes have a lower BMR than ordinary persons due to decreased muscle mass
B-	Athletes have a higher BMR than ordinary persons, by about 10%, due to increased muscle mass
C-	Physical activity has no effect on BMR
D-	Only aerobic (not resistance) training affects BMR

56-	**Regarding the effect of dietetic habits on BMR: (M3WAN - Core)**
A-	A long-term protein-rich diet increases BMR by about 10% more thana carbohydrate-rich diet
B-	A carbohydrate-rich diet increases BMR more thana protein-rich diet
C-	Diet composition has no effect on BMR
D-	Only fat-rich diets affect BMR

57-	**Regarding the effect of climate on BMR: (M3WAN - Core)**
A-	BMR is higher in hot zones than in cold/temperate zones
B-	BMR is higher in cold/temperate zones (e'g', in Eskimos) than in hot zones, by about 10%
C-	Climate has no effect on BMR
D-	BMR is identical in all climates

58-	**Regarding the effect of pregnancy on BMR: (Important M3WAN - Core)**
A-	BMR decreases progressively during pregnancy
B-	BMR remains unchanged throughout pregnancy
C-	BMR increases gradually during pregnancy, becoming doubled by the end of pregnancy, due to increased thyroxine, cortisol, and sex hormones plus added fetal metabolism
D-	BMR only changes in the first trimester

59-	**Regarding the effect of sleep on BMR: (M3WAN - Core)**		
A-	Sleep decreases BMR by about 10%	B-	Sleep increases BMR by about 10%
C-	Sleep has no effect on BMR	D-	Sleep decreases BMR by about 50%

60-	**Among pathological factors, hyperthyroidism increases BMR by up to 100% or higher, mainly because: (Important M3WAN - Core)**
A-	Thyroxine is the most powerful stimulant of cell metabolism
B-	Thyroxine has no metabolic effect
C-	Hyperthyroidism decreases oxygen consumption
D-	Hyperthyroidism is unrelated to metabolic rate

61-	**Regarding the effect of fever on BMR: (Important M3WAN - Core)**
A-	A1°Crise in body temperature decreases BMR by 13%
B-	A1°Crise in body temperature increases BMR by about 13%
C-	Fever has no effect on BMR
D-	A1°Crise in body temperature doubles the BMR

62-	**Hypothyroidism and hypothermia both tend to: (M3WAN - Core)**		
A-	Increase the BMR	B-	Decrease the BMR
C-	Have no effect on the BMR	D-	Cause identical effects on RQ but not BMR

63-	**Specific dynamic action (SDA) of food is defined as: (Important M3WAN - Core)**
A-	The power of food to stimulate the metabolic rate (the thermic effect of food)
B-	The caloric value of food measured by bomb calorimeter
C-	The rate of gastric emptying
D-	The biological value of dietary protein

64-	**Regarding the time course of specific dynamic action: (M3WAN - Core)**
A-	It starts immediately and ends within 10 minutes
B-	The metabolic rate starts to increase 1 hour after taking food, reaches a maximum after 3-4 hours, and returns to basal levels 12-18 hours after food intake
C-	It has no defined time course
D-	It occurs only during sleep

65-	**Among the three main food types, the highest specific dynamic action is produced by: (Important M3WAN - Core)**		
A-	Fat (4%)	B-	Carbohydrate (6%)
C-	Protein (30%)	D-	All three produce identical SDA

66-	**Specific dynamic action is influenced by external temperature such that: (Important M3WAN - Core)**
A-	SDA is maximal at an external temperature of about 33°C, and disappears at about 5°Cbecause the heat is retained to maintain body temperature
B-	SDA is unaffected by external temperature
C-	SDA disappears at high external temperatures and is maximal in the cold
D-	SDA only occurs at freezing temperatures

67-	**Regarding the role of the liver in specific dynamic action: (Important M3WAN - Core)**
A-	The liver plays no role in SDA
B-	The liver is essential for the SDA of protein, performing oxidative deamination of amino acids; SDA of protein is markedly reduced or absent if the liver is removed
C-	The liver is essential for the SDA of fat only
D-	The liver only affects the SDA of carbohydrate

68-	**Regarding specific dynamic action and the biological value of protein: (M3WAN - Hard)**
A-	A low biological value protein produces a much lower SDA thana high biological value protein
B-	SDA is not related to the biological value of protein; a low biological value protein produces the same SDA as a high biological value protein
C-	SDA is entirely determined by biological value
D-	Only high biological value proteins produce any SDA at all

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	B	C	A	D	A	C	A
11	12	13	14	15	16	17	18	19	20
C	B	C	B	C	A	B	B	D	C
21	22	23	24	25	26	27	28	29	30
D	C	C	B	A	C	B	B	A	D
31	32	33	34	35	36	37	38	39	40
B	A	B	A	A	A	A	A	A	B
41	42	43	44	45	46	47	48	49	50
C	A	B	A	B	A	B	C	A	D
51	52	53	54	55	56	57	58	59	60
B	A	C	A	B	A	B	C	A	A
61	62	63	64	65	66	67	68	69	70
B	B	A	B	C	A	B	B	-	-

Food intake

Pre Exams & Formative

1-	**POMC neurons in arcuate nucleus of hypothalamus releasesa'MSH which? (FINAL 31)**
A-	Causes darkening of the skin by increasing melanin pigment
B-	Decreases food intake
C-	Decreases the energy expenditure
D-	Decreases the BMR

2-	**Leptin hormone secreted by? (FINAL 31, FINAL 32, 2ND TRIAL 35, Formative 36)**						
A-	Hypothalamus	B-	Stomach	C-	Adipose tissue	D-	Pituitary gland

3-	**The following factors lead to decrease food intake, EXCEPT? (FINAL 31)**
A-	Receptors in the mouth and esophagus which monitor food intake
B-	Receptors which are stimulated by distension of the stomach
C-	GIT hormones
D-	Cold receptors

4-	**NeuropeptideY released from hypothalamus neurons and causes? (FINAL 32)**		
A-	Increase food intake	B-	Decrease food intake
C-	Increase energy expenditure	D-	Decrease body weight

5-	**Increase blood level of which of the following increases the food intake? (2ND TRIAL 32, 2ND TRIAL 35)**						
A-	Leptin	B-	Ghrelin	C-	Peptide YY	D-	Insulin

6-	**Which one of the following hormones does not induce satiety? (2ND TRIAL 35)**						
A-	Ghrelin	B-	Insulin	C-	Cholecystokinin	D-	Peptide YY

7-	**The hormone Leptin do its effects by: (FINAL SNU 1)**
A-	Inhibiting the Arcuate nucleus.
B-	Acting on and increasing the activity of POMC neurons.
C-	Decreasing the metabolic rate.
D-	Stimulating the release of Ghrelin.

8-	**The most important long-term inhibitor. (FINAL 37)**		
A-	Level of glucose	B-	The level of fatty acids

9-	All of the following is true about pro-opiomelanocortin except? (FINAL 37)		
A-	Cover melanoreceptors on paraventricular	B-	Arise from Arcuate nuclei
C-	Secret melano stimulating hormone	D-	Increase food intake

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
B	C	D	A	B	A	B	B	D	-

Department Qs

1-	Concerning the appetite, all the following is true except: (SNU Bank, Dr. Mohamed Ashraf)		
A-	It is affected by certain hypothalamic areas.	B-	It is completely lost after gastrectomy.
C-	It is reduced if the stomach is over distended.	D-	It increases as the blood glucose level falls.
E-	It increases on exposure to cold climates.		

2-	which of the following isa neurotransmitter that stimulates the feeding center: (SNU Bank, Dr. Mohamed Ashraf)						
A-	peptide YY	B-	insulin	C-	neuropeptideY.	D-	MSH

3-	Leptin isa hormone released from (SNU Bank, Dr. Mohamed Ashraf)						
A-	empty stomach	B-	full stomach	C-	hypothalamus	D-	adipose tissue

4-	which of the following factors decreases the food intake (SNU Bank, Dr. Mohamed Ashraf)		
A-	decreased blood glucose level	B-	exposure to cold weather
C-	increase gut peptide as CCK	D-	increase gut peptide as gherlin

5-	POMC neurons release a MSH which? (SNU Bank)		
A-	Causes skin darkening	B-	Decreases food intake
C-	Decreases energy expenditure	D-	Decreases BMR

6-	Leptin hormone is secreted by? (SNU Bank)						
A-	Hypothalamus	B-	Stomach	C-	Adipose tissue	D-	Pituitary gland

7-	Factors that decrease food intake EXCEPT? (SNU Bank)
A-	Receptors in mouth/esophagus monitoring food
B-	Receptors stimulated by stomach distension
C-	GIT hormones
D-	Cold receptors

8-	NeuropeptideY (NPY) released from hypothalamus causes? (SNU Bank)		
A-	Increase food intake	B-	Decrease food intake
C-	Increase energy expenditure	D-	Decrease body weight

9-	Leptin is secreted by? (SNU Bank)						
A-	Hypothalamus	B-	Adipose tissue	C-	Stomach	D-	Colon

10-	Increase blood level of which of the following increases the food intake? (SNU Bank)						
A-	Leptin	B-	Ghrelin	C-	Peptide YY	D-	Insulin

11-	Leptin isa hormone produced by? (SNU Bank)						
A-	Liver	B-	Adipose tissue	C-	Stomach	D-	Pancreas

12-	Which one of the following hormones does NOT induce satiety? (SNU Bank)						
A-	Ghrelin	B-	Insulin	C-	Cholecystokinin	D-	Peptide YY

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	C	D	C	B	C	D	A	B	B
11	12	13	14	15	16	17	18	19	20
B	A	-	-	-	-	-	-	-	-

M3WAN Bank

1-	Control of food intake is achieved by specific centers located in the: (M3WAN - Core)		
A-	Cerebellum only	B-	Brainstem only
C-	Spinal cord only	D-	Hypothalamus and limbic system

2-	The feeding center is located in the: (Important M3WAN - Core)		
A-	Ventromedial hypothalamus	B-	Paraventricular nucleus
C-	Lateral hypothalamus	D-	Preoptic nucleus

3-	Stimulation of the feeding center results in: (Important M3WAN - Core)		
A-	Anorexia	B-	Increased appetite and food intake
C-	Decreased appetite and food intake	D-	No change in eating behavior

4-	Destruction of the feeding center results in: (M3WAN - Core)		
A-	Anorexia	B-	Hyperphagia

C-	No change in appetite	D-	Increased thirst only

5-	**The satiety center is located in the: (Important M3WAN - Core)**		
A-	Lateral hypothalamus	B-	Posterior hypothalamus
C-	Ventromedial hypothalamus	D-	Dorsomedial nucleus

6-	**Stimulation of the satiety center results in: (Important M3WAN - Core)**		
A-	Increased appetite and food intake	B-	Anorexia only, with no effect on appetite
C-	Excessive thirst	D-	Decreased appetite and food intake

7-	**Destruction of the satiety center results in: (M3WAN - Core)**		
A-	Anorexia	B-	Hyperphagia
C-	No change in food intake	D-	Increased energy expenditure

8-	**Under normal conditions, the feeding center is: (M3WAN - Core)**
A-	Continuously active and inactivated only when the satiety center is stimulated
B-	Continuously inactive unless directly stimulated
C-	Active only during sleep
D-	Active only after gastric distension

9-	**Lesions of the paraventricular nucleus often cause: (M3WAN - Core)**		
A-	Anorexia	B-	Depressed eating behavior
C-	No change in eating behavior	D-	Excessive eating

10-	**Lesions of the dorsomedial nucleus usually: (M3WAN - Core)**		
A-	Cause excessive eating	B-	Have no effect on eating behavior
C-	Depress eating behavior	D-	Cause increased energy expenditure only

11-	**The arcuate nuclei of the hypothalamus serve as the site where: (M3WAN - Core)**
A-	Only thermoregulatory signals converge
B-	Multiple GIT hormones and adipose tissue signals converge to regulate food intake and energy expenditure
C-	Only visual signals related to food are processed
D-	Motor commands for chewing are generated

12-	**The arcuate nuclei contain two main types of neurons concerned with appetite and energy expenditure regulation, namely: (M3WAN - Core)**		
A-	POMC neurons and NPY/AgRP neurons	B-	Muscarinic and nicotinic neurons
C-	Alpha and beta neurons	D-	Sensory and motor neurons

13-	**Pro-opiomelanocortin (POMC) neurons in the arcuate nucleus produce: (Important M3WAN - Core)**
A-	NeuropeptideY and Agouti-related protein
B-	Ghrelin and cholecystokinin
C-	Insulin and glucagon
D-	Alpha-melanocyte-stimulating hormone (α-MSH) and CART

14-	**Activation of POMC neurons in the arcuate nucleus: (M3WAN - Core)**
A-	Increases food intake and decreases energy expenditure
B-	Has no effect on food intake or energy expenditure
C-	Decreases food intake and increases energy expenditure
D-	Only affects skin pigmentation, with no metabolic effect

15-	**POMC neurons are stimulated by: (M3WAN - Core)**		
A-	Satiety signals, such as leptin	B-	Hunger signals, such as ghrelin
C-	Only mechanical distension of the stomach	D-	Cold exposure exclusively

16-	**NPY/AgRP neurons in the arcuate nucleus produce the orexigenic substances: (Important M3WAN - Core)**
A-	α-MSH and CART
B-	NeuropeptideY (NPY) and Agouti'related protein (AgRP)
C-	Insulin and calcitonin
D-	Cholecystokinin and peptide YY

17-	**Activation of NPY/AgRP neurons: (M3WAN - Core)**
A-	Decreases food intake and increases energy expenditure
B-	Has no effect on appetite
C-	Only affects thermoregulation
D-	Increases food intake and reduces energy expenditure

18-	**NPY/AgRP neurons are activated by: (M3WAN - Core)**		
A-	Satiety signals, such as leptin	B-	Only cold exposure
C-	Hunger signals, such as ghrelin	D-	Only high blood glucose

19-	**Alpha-MSH released by POMC neurons acts on melanocortin receptors (MCR-3 and MCR-4) located on the: (M3WAN - Core)**
A-	Lateral hypothalamus
B-	Paraventricular nuclei, reducing food intake and increasing energy expenditure
C-	Preoptic nucleus, regulating body temperature
D-	Suprachiasmatic nucleus, regulating circadian rhythm

20-	**Short-term regulation of food intake is primarily concerned with: (M3WAN - Core)**
A-	Preventing overeating at each meal
B-	Maintaining normal long-term energy stores in the body
C-	Regulating body temperature
D-	Controlling long-term body weight over months

21-	**Long-term regulation of food intake is primarily concerned with: (M3WAN - Core)**
A-	Preventing overeating at a single meal
B-	Metering food as it passes through the buccal cavity
C-	Immediate gastric distension signals
D-	Maintenance of normal quantities of energy stores in the body

22-	**GIT receptors involved in short-term regulation of food intake are present in the buccal cavity, esophagus, stomach, and intestine; stimulation of these receptors (e'g', by gastric distension) acts via: (M3WAN - Core)**
A-	Direct action on the adrenal medulla
B-	The optic nerve
C-	Vagal afferents, which stimulate the satiety center and decrease food intake
D-	Sympathetic efferents to the pancreas only

23-	**Cholecystokinin (CCK), released from I cells in the duodenal mucosa: (Important M3WAN - Core)**
A-	Increases appetite and food intake
B-	Decreases appetite and food intake, but is short-acting and therefore not useful in obesity treatment
C-	Hasa long duration of action, making it ideal for obesity treatment
D-	Is released only from the pancreas

24-	**Cholecystokinin (CCK) decreases food intake mainly by: (M3WAN - Core)**
A-	Exciting sensory endings in the duodenum that travel in the vagus nerve
B-	Directly crossing the blood-brain barrier to inhibit the feeding center
C-	Stimulating the lateral hypothalamus directly
D-	Increasing gastric emptying rate

25-	Compared with CCK, peptide YY (PYY), released mainly from the ileum and colon: (Important M3WAN - Core)
A-	Hasa shorter duration of action than CCK
B-	Increases appetite after eating
C-	Hasa longer duration of action than CCK, signaling fullness and suppressing appetite
D-	Is released only during fasting

26-	Ghrelin, released from the gastric mucosa and pancreas mainly during fasting: (Important M3WAN - Core)		
A-	Decreases appetite and food intake	B-	Has no effect on feeding behavior
C-	Is released only after a large meal	D-	Stimulates feeding

27-	Hyperglycemia tends to: (M3WAN - Core)		
A-	Stimulate the feeding center	B-	Stimulate the satiety center
C-	Have no effect on either center	D-	Stimulate ghrelin release

28-	Despite having a high blood glucose level, diabetics often show hyperphagia because: (Important M3WAN - Core)
A-	The satiety center requires insulin for glucose transport into its cells, so it cannot be adequately stimulated
B-	The feeding center is destroyed in diabetes
C-	Blood glucose has no relation to appetite regulation
D-	Ghrelin secretion is permanently suppressed in diabetes

29-	Hypoglycemia tends to: (M3WAN - Core)
A-	Stimulate the satiety center
B-	Have no effect on the feeding or satiety centers
C-	Stimulate leptin release
D-	Stimulate the feeding center

30-	Rising amino acid and fatty acid levels in the blood tend to inhibit feeding by: (M3WAN - Core)		
A-	Directly damaging the feeding center	B-	Stimulating ghrelin release
C-	Negative feedback	D-	Stimulating the dorsomedial nucleus only

31-	Among long-term regulators of food intake, the most important long-term inhibitor of feeding is the level of: (Important M3WAN - Core)						
A-	Blood glucose	B-	Fatty acids	C-	Blood calcium	D-	Blood urea

32-	**Exposure to cold tends to: (Important M3WAN - Core)**		
A-	Stimulate the feeding center	B-	Stimulate the satiety center
C-	Have no effect on food intake regulation	D-	Suppress ghrelin secretion

33-	**Exposure to heat tends to: (Important M3WAN - Core)**		
A-	Stimulate the feeding center	B-	Have no effect on food intake regulation
C-	Suppress leptin secretion	D-	Stimulate the satiety center

34-	**Regarding hormonal effects on food intake, insulin and calcitonin: (M3WAN - Core)**
A-	Increase appetite and food intake
B-	Have no central effect on appetite
C-	Decrease appetite and food intake by acting on central receptors in the brain
D-	Act only peripherally, with no CNS effect

35-	**Regarding hormonal effects on food intake, glucagon: (M3WAN - Core)**		
A-	Increases appetite and food intake	B-	Decreases appetite and food intake
C-	Has no effect on appetite	D-	Acts identically to insulin

36-	**Leptin a hormone important in long-term regulation of food intake, is secreted mainly by: (Important M3WAN - Core)**		
A-	The hypothalamus	B-	Adipose tissue
C-	The stomach	D-	The pancreas

37-	**Leptin crosses the blood-brain barrier by facilitated diffusion and acts on POMC and AgRP/NPY neurons of the arcuate and paraventricular nuclei to: (Important M3WAN - Core)**
A-	Stimulate the feeding center and decrease energy expenditure
B-	Have no effect on energy balance
C-	Stimulate NPY and AgRP release directly
D-	Inhibit the feeding center, helping control body weight and increase energy expenditure

38-	**Among drugs affecting food intake, amphetamine acts mainly by: (M3WAN - Core)**		
A-	Stimulating the feeding center	B-	Blocking leptin receptors
C-	Stimulating the satiety center	D-	Increasing ghrelin secretion

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	B	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	A	C	D	B	A	D	C
31	32	33	34	35	36	37	38	39	40
B	A	D	C	A	B	D	C	-	-

(النَّبِيُّ أَوْلَى بِالْمُؤْمِنِينَ مِنْ أَنْفُسِهِمْ)

أقربُ ما للإنسانِ، وأولى ما له نفسُه، فالرَّسولُ ﷺ أولى به من نفسِه.

لأنَّه عليه الصَّلاةُ والسَّلامُ، بذلَ لهم من النُّصحِ، والشَّفقةِ، والرَّأفةِ، ما كان به أرحمَ الخلقِ، وأرأفهم، فرسولُ اللهِ ﷺ، أعظمُ الخلقِ منَّةً عليهم، من كلِّ أحدٍ، فإنَّه لم يصلِ إليهم مثقالُ ذرَّةٍ من الخيرِ، ولا اندفعَ عنهم مثقالُ ذرَّةٍ من الشرِّ، إلَّا على يديه وبسببه.

فلذلك، وجبَ عليهم إذا تعارضَ مرادُ النَّفسِ، أو مرادُ أحدٍ من النَّاسِ، مع مرادِ الرَّسولِ ﷺ، أن يُقدِّمَ مرادَ الرَّسولِ ﷺ، وأن لا يُعارضَ قولَ الرَّسولِ ﷺ، بقولِ أحدٍ، كائنًا من كان، وأن يفدُوه بأنفسِهم وأموالِهم وأولادِهم، ويُقدِّموا محبَّتَه على الخلقِ كلِّهم.»

عبدُ الرَّحمنِ السَّعدي -رحمَهُ اللهُ.-

[تفسيرُ السَّعديِّ || ١٣٧٣]

Obesity

Pre Exams & Formative

1-	Regarding Obesity all the following are true EXCEPT? (FINAL 31)		
A-	Is usually caused by hyperphagia	B-	Is predisposes to cardiovascular diseases
C-	In childhood it is difficult to treat	D-	Body Mass Index is less than 25

2-	All about obesity is correct, EXCEPT? (FINAL 31)
A-	It is associated with body mass index above 30
B-	Isa cause of diabetes mellitus
C-	In parents is associated with obesity in their children
D-	Is usually due to endocrinal disorder, mainly hyperthyroidism

3-	The ideal treatment for morbid obesity is? (FINAL 31)		
A-	Physical exercises	B-	Drugs decreasing appetite
C-	Diet control	D-	Bariatric surgery

4-	Obesity is considered when BMI increases above? (FINAL 32, 2ND TRIAL 32)					
A-	20	B-	25	C-	30	D-	35

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
D	D	D	C	-	-	-	-	-	-

Department Qs

1-	About obesity, all the following is true except: (SNU Bank, Dr. Mohamed Ashraf)
A-	It is the commonest nutritional disorder.
B-	It is associated with a fall in the body's specific gravity.
C-	It is produced by deposition of excess protein in the body.
D-	It occurs as a result of +ve energy balance often due to hyperphagia.
E-	Psychogenic. genetic and hypothalamic abnormalities may be causative factors.

2-	Which of the following statements about obesity is wrong? (SNU Bank, Dr. Mohamed Ashraf)
A-	Obesity may be hypertrophic or hyperplastic-hypertrophic.
B-	Some cases of obesity are due to resetting of the hypothalamic appestat toa higher level than normal.

C-	Obesity is present when more than 20% of the body weight is due to fat in males and more than 25% in females.
D-	Regular exercises and slimming food (containing no fat and low carbohydrate) usually reduce the body weight.
E-	The use of diuretics and thyroxine for reducing the body weight is not advisable due to their harmful effects
F-	Amphetamine is not useful in its treatment since it increases the appetite.

3-	**obesity can be diagnosed by estimation of (SNU Bank, Dr. Mohamed Ashraf)**						
A-	BMR	B-	BMI	C-	BMW	D-	RQ

4-	**among the complications of obesity: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	hypotension	B-	diabetes mellitus
C-	hypolipidemia	D-	rapid healing of wounds

5-	**the first choice in management of obesity is (SNU Bank, Dr. Mohamed Ashraf)**		
A-	correction of dietary habits and exercise	B-	anorexigenic drugs
C-	gastric bypass surgery	D-	liposuction

6-	**the person is considered overweight when his (SNU Bank, Dr. Mohamed Ashraf)**		
A-	weight is higher than 70 kg	B-	basal metabolic rate is higher than 40C
C-	body mass index between 25-30	D-	respiratory quotient higher than one

7-	**the person is considered obese when his BMI is (SNU Bank, Dr. Mohamed Ashraf)**		
A-	less than 18	B-	between 18 and 25
C-	between 25 and 30	D-	more than 30

8-	**Regarding obesity, all true EXCEPT? (SNU Bank)**		
A-	Usually caused by hyperphagia	B-	Predisposes to cardiovascular diseases
C-	In childhood is difficult to treat	D-	BMI is less than 25

9-	**All about obesity correct EXCEPT? (SNU Bank)**
A-	Associated with BMI above 30
B-	Isa cause of diabetes mellitus
C-	In parents associated with obesity in children
D-	Usually due to endocrinal disorder, mainly hyperthyroidism

10-	**Ideal treatment for morbid obesity? (SNU Bank)**		
A-	Physical exercises	B-	Drugs decreasing appetite

C-	Diet control	D-	Bariatric surgery

11-	**Obesity is considered when BMI increases above? (SNU Bank)**						
A-	20	B-	25	C-	30	D-	35

Answers - Department Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	**-**	**B**	**B**	**A**	**C**	**D**	**D**	**D**	**D**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

M3WAN Bank

1-	**Obesity is best defined as: (M3WAN - Core)**
A-	A BMI below 18.5
B-	Deficiency of fat in the storage areas of the body
C-	A state of chronic undernutrition
D-	Deposition of an excessive amount of fat in the storage areas of the body

2-	**Functions of fat tissue include all of the following EXCEPT: (M3WAN - Core)**		
A-	Giving the body its shape	B-	Protecting the viscera from external trauma
C-	Direct conduction of nerve impulses	D-	Acting as an insulator and energy reserve

3-	**Fat tissue is now considered a major endocrine organ because it secretes substances such as: (M3WAN - Core)**
A-	Insulin and glucagon
B-	Leptin, adiponectin, resistin, and other cytokines
C-	Thyroxine and calcitonin
D-	Acetylcholine and noradrenaline

4-	**Obesity occurs mainly when: (M3WAN - Core)**		
A-	Energy input exceeds energy output	B-	Energy output exceeds energy input
C-	Energy input equals energy output exactly	D-	Protein intake exceeds carbohydrate intake

5-	**Obesity is said to be present when body fat reaches or exceeds approximately: (Important M3WAN - Core)**						
A-	10% in males	B-	15% in females	C-	25% in males	D-	45% in females

6-	Obesity is said to be present when body fat reaches or exceeds approximately: (Important M3WAN - Core)						
A-	15% in females	B-	20% in males	C-	10% in females	D-	35% in females

7-	Obesity is estimated clinically using the body mass index (BMI), calculated as: (M3WAN - Core)		
A-	Weight (kg) multiplied by height (m)	B-	Weight (kg) divided by height squared (m2)
C-	Height (m) divided by weight (kg)	D-	Weight (kg) divided by age (years)

8-	ABMI of 25 to 29.9 is classified as: (Important M3WAN - Core)						
A-	Overweight	B-	Obesity	C-	Underweight	D-	Normal weight

9-	ABMI of 30 or more is classified as: (Important M3WAN - Core)						
A-	Overweight only	B-	Normal weight	C-	Underweight	D-	Obesity

10-	Hyperplastic obesity is characterized by: (M3WAN - Core)		
A-	An increase in the size of existing adipocytes	B-	No change in adipocyte number or size
C-	An increase in the number of adipocytes	D-	A decrease in the number of adipocytes

11-	Hyperplastic obesity mainly occurs in: (M3WAN - Core)
A-	Adults, since adipocyte number can increase at any age
B-	Children, since adipocyte number increases mainly during childhood
C-	The elderly exclusively
D-	Pregnant women exclusively

12-	Hypertrophic obesity is characterized by: (M3WAN - Core)		
A-	An increase in the size of existing adipocytes	B-	An increase in the number of adipocytes
C-	No change in adipocyte size	D-	A decrease in adipocyte size

13-	Hypertrophic obesity mainly occurs in: (M3WAN - Core)		
A-	Children exclusively	B-	Newborn infants exclusively
C-	Fetal life exclusively	D-	Adults

14-	The most common cause of obesity is: (Important M3WAN - Core)		
A-	Hypothyroidism	B-	Cushing's syndrome
C-	Hyperphagia	D-	Genetic abnormality of neurotransmitters

15-	Hyperphagia leading to obesity may result from: (M3WAN - Core)
A-	Psychological interference with the normal function of the hypothalamic feeding center

B-	Complete destruction of the feeding center
C-	Complete destruction of the satiety center only
D-	Increased leptin sensitivity

16-	**A sedentary lifestyle contributes to obesity mainly through: (M3WAN - Core)**
A-	Increased food intake with normal energy expenditure
B-	Decreased energy expenditure with food intake remaining normal (relative hyperphagia)
C-	Decreased food intake with normal energy expenditure
D-	Increased energy expenditure with decreased food intake

17-	**Among the endocrinal causes of obesity is: (M3WAN - Core)**		
A-	Hyperthyroidism	B-	Addison's disease
C-	Hypoinsulinism	D-	Hypothyroidism (myxoedema)

18-	**Among the endocrinal causes of obesity is: (M3WAN - Core)**		
A-	Type 1 diabetes mellitus	B-	Hypocorticism
C-	Hyperinsulinism	D-	Hypopituitarism

19-	**Among the endocrinal causes of obesity is: (M3WAN - Core)**		
A-	Addison's disease	B-	Cushing's syndrome
C-	Hypothyroidism only in adults	D-	Diabetes insipidus

20-	**Frohlich's syndrome (adipose-genital syndrome) causes obesity mainly by: (Important M3WAN - Core)**
A-	Affecting the satiety center in the hypothalamus, leading to hyperphagia
B-	Directly increasing the number of adipocytes in adulthood
C-	Increasing thyroid hormone secretion
D-	Decreasing cortisol secretion

21-	**Obesity resulting from underlying emotional disturbance is classified as: (M3WAN - Core)**		
A-	Hyperplastic obesity	B-	Hypertrophic obesity
C-	Endocrinal obesity	D-	Psychogenic obesity

22-	**Genetic obesity is thought to result from: (M3WAN - Core)**
A-	An abnormality confined to adipocyte number only
B-	Excess dietary carbohydrate intake exclusively
C-	An abnormality in the feeding center or its neurotransmitters
D-	A defect limited to insulin receptors

23-	Complications of obesity include all of the following EXCEPT: (M3WAN - Core)		
A-	Accelerated atherosclerosis	B-	Improved respiratory function
C-	Type II diabetes mellitus	D-	Osteoarthritis

24-	Obesity isa recognized risk factor for the development of: (Important M3WAN - Core)		
A-	Type II diabetes mellitus	B-	Type I diabetes mellitus exclusively
C-	Hyperthyroidism	D-	Addison's disease

25-	Obesity is associated with an increased risk of: (M3WAN - Core)		
A-	Improved joint mobility	B-	Decreased cardiovascular disease risk
C-	Cancer	D-	Decreased risk of shortness of breath

26-	Musculoskeletal complications of obesity include: (M3WAN - Core)
A-	Improved joint cartilage regeneration
B-	Decreased joint loading
C-	Reduced risk of osteoarthritis
D-	Osteoarthritis, due to increased mechanical loading on joints

27-	The general principles of obesity treatment include: (M3WAN - Core)
A-	Increasing energy intake and decreasing energy expenditure
B-	Decreasing energy intake and increasing energy expenditure
C-	Increasing both energy intake and energy expenditure equally
D-	Complete elimination of dietary fat only

28-	Amphetamine is used in obesity treatment mainly because it: (M3WAN - Core)		
A-	Decreases appetite and hunger pains	B-	Increases fat absorption from the gut
C-	Directly reduces adipocyte number	D-	Increases insulin secretion

29-	Orlistat is used in obesity treatment mainly because it: (Important M3WAN - Core)		
A-	Directly suppresses the feeding center	B-	Increases gastric emptying
C-	Stimulates leptin secretion	D-	Alters lipid absorption by the gut

30-	Surgical treatment of obesity may involve: (M3WAN - Core)
A-	Removal of the entire pancreas
B-	Destruction of the satiety center
C-	Reducing the size of the stomach reservoir and/or bypassing it altogether
D-	Removal of all visceral adipose tissue

31-	**Regarding obesity in childhood: (Important M3WAN - Core)**
A-	It is generally easier to treat than adult obesity
B-	It is generally difficult to treat
C-	It never involves an increase in adipocyte number
D-	It is always due to endocrinal causes

32-	**Obesity predisposes to: (M3WAN - Core)**		
A-	Cardiovascular disease	B-	Decreased risk of type II diabetes mellitus
C-	Improved lung function	D-	Decreased mechanical joint stress

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	B	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	A	C	D	B	A	D	C
31	32	33	34	35	36	37	38	39	40
B	A	-	-	-	-	-	-	-	-

تم بفضل الله ، هذا وما كان من توفيق فمن الله وحده وما كان من خطأ أو زلل فمن أنفسنا ومن الشيطان ، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا على طاعته .

نسألكم الدعاء لكل من سبقنا من إخواننا إلى الله أن يتغمدهم بواسع رحمته وأن يرفع قدرهم ويتجاوز عن سيئاتهم وأن يجمعنا بهم في جناته .